

# Informed Trading Decisions via Social Network Analysis: A Graph Learning Approach



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## Abstract

Cryptocurrency markets are highly sensitive to public sentiment, yet existing approaches have not fully exploited social media data to understand how that sentiment relates to financial behaviour and prediction. For instance, most existing approaches aggregate sentiment across large populations and therefore overlook individual forecasting ability of users. This project investigates whether high-predictive users can be identified from their behaviour and position within an online discussion network, and analysis is conducted over the posts and comments collected from the subreddit *r/CryptoMarkets* submitted between June 2024 and June 2025. Due to the difficulty in extracting a structured forecast schema, language models were implemented, compared and validated. Forecast statements were extracted from informal Reddit discourse, aligned with subsequent market movements, and used to assign users a label based on their predictive performance. These users were then represented as nodes in a weighted social interaction graph, with node features capturing behavioural, textual, and structural characteristics.

The problem was formulated as a semi-supervised node classification task under class imbalance, and several baseline and graph neural networks were compared, including MLP, LP, GCN, GAT, GraphSAGE, H<sub>2</sub>GCN and MixHop. Among these, GraphSAGE achieved the strongest threshold-independent performance, despite the severe class imbalance and heterophily, as it achieved an average precision of 0.140 and an AUROC of 0.632, corresponding to a 60.9% improvement in average precision over the random baseline and a clear and consistent improvement over feature-only baselines. Ablation and perturbation analyses further showed that predictive performance is driven by a combination of behavioural engagement patterns and influence-based structural signals, rather than by raw social network activity alone.

These findings indicate that predictive capability is not uniformly distributed across the community, but is associated with measurable and complex characteristics of user participation and network position. More broadly, the project demonstrates how graph learning can be used to identify informative participants in large-scale online financial discussions.

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# 1 Introduction

The global cryptocurrency market is uniquely sensitive to public sentiment. Unlike traditional financial assets, cryptocurrencies are not backed by government guarantees or company fundamentals, making collective online opinion a primary driver of price movements. This dependence on sentiment makes social media an especially informative data source for understanding and anticipating cryptocurrency market behaviour. Reddit is particularly well-suited due to its topic-oriented community structure, long-form posting format, and high user engagement; producing substantive and searchable financial discourse [1]. The subreddit *r/CryptoMarkets* alone attracts 1.9 million members and over 3,800 weekly contributions, making it a rich environment for studying how online discussion relates to market dynamics.

Prior work has established that Reddit sentiment correlates with cryptocurrency price movements, with certain subreddits acting as early indicators of market trends, especially for Bitcoin and Ethereum [2, 3]. However, a critical limitation pervades the existing literature. Predictive signals are typically derived by aggregating sentiment across entire subreddits or platforms, treating every user’s contribution as equally informative [2, 4]. This assumption is unlikely to hold in practice. Online communities are heterogeneous as some participants post consistently accurate directional forecasts while others contribute noise, speculation, or herd-driven commentary. Wooley explicitly suggests that predictive information may originate from only a subset of community participants [3], and studies of platforms such as *r/WallStreetBets* show that the presence of highly active users and coordinated behaviour can distort aggregate outcomes [4]. Yet no existing work has developed a principled, scalable framework for identifying which individual users provide consistently reliable forecasts. This gap is the central motivation for the present work.

Identifying such users is non-trivial for two reasons. First, cryptocurrency forecasts in informal social media text are implicit, hedged, and colloquial, making automated extraction unreliable with simple heuristics. Second, even if forecasting accuracy can be measured, it is unclear how to identify high-predictive users before sufficient forecast history has been accumulated, in order to act on their contributions in real time. Addressing both challenges requires combining natural language processing, market data alignment, and a model that can generalise from limited labelled examples to the broader user population, motivating a semi-supervised formulation that exploits the structure of unlabelled users alongside the labelled subset.

This project proposes that the social interaction network of *r/CryptoMarkets* carries structural information relevant to this task. Interaction on online communities are not random as interaction patterns reflect shared interests, expertise and influence. Research in social network analysis has demonstrated

that structural position within a network, captured through properties such as community membership, centrality and neighbourhood composition, is informative about node-level attributes [5]. If high-predictive users exhibit distinctive interaction patterns or occupy identifiable structural positions, then a model that incorporates both user behaviour and graph topology should outperform one that treats users independently. Therefore, this problem is designed as a semi-supervised transductive node classification on a weighted social interaction graph, where labelled users provide the training signal, unlabelled users are observed as part of the graph during training, and graph structure propagates information from labelled to unlabelled nodes.

This project makes three principal contributions. First, a scalable pipeline is developed to extract cryptocurrency forecasts from informal Reddit discourse, evaluate them against realised market movements, and assign users composite predictive scores via confidence-weighted accuracy and Bayesian shrinkage. This addresses a gap in the literature by providing a principled method in quantifying the forecasting ability of individual social media users, which is a prerequisite for any downstream classification task. Second, the identification of high-predictive users was implemented with semi-supervised graph machine-learning, integrating behavioural, textual, and structural node features. A comparative evaluation of seven graph machine learning architectures, specifically MLP, Label Propagation, GCN, GAT, GraphSAGE, H<sub>2</sub>GCN, and MixHop, demonstrates that GraphSAGE achieves the strongest overall performance, with a 60.9% improvement in average precision and a 26.9% improvement in AUROC over a random baseline, confirming that combining node attributes with neighbourhood aggregation yields a consistent advantage over both feature-only and structure-only approaches. Third, ablation and topological perturbation analyses identify the structural and behavioural features that are associated with model performance, providing interpretable insight into what distinguishes high-predictive users within the network. These findings contribute to a broader understanding of how expertise is distributed in large-scale online financial communities, with implications for automated signal filtering, expert identification and information quality assessment across social platform.

The remainder of this report is structured as follows. Chapter 2 reviews related literature and relevant graph theory. Chapters 3 to 5 detail the problem formulation, data pipeline, and network construction. Chapters 6 to 8 describe the models, results, and discussion. Chapter 9 concludes with directions for future work.

## 2 Background and Related Work

In this chapter, an introduction to the current relevant work and basic graph principles is given.

### 2.1 Related Work

#### 2.1.1 Reddit-based Cryptocurrency Forecasting

Research in financial market prediction relies on historical prices, trading volumes, and technical indicators derived from market microstructure. In cryptocurrency markets, these approaches are often limited due to high volatility, speculative dynamics, and the lack of widely accepted fundamental valuation models. This has motivated increasing interest in alternative data sources, particularly social media, as a means of capturing market sentiment and collective expectations.

Early work in this area focused on extracting sentiment signals from financial news and social media content and examining their relationship with subsequent market movements. Studies show that sentiment embedded in online discourse can correlate with asset price changes and, in some cases, provide predictive information beyond traditional indicators [6]. In practice, such approaches typically consolidate sentiment across large populations of users or documents, treating individual contributions as interchangeable when constructing predictive signals.

Subsequent research has examined whether aggregated online opinions reflect collective intelligence or are instead shaped by herd behaviour and participation biases. In the context of the subreddit *r/WallStreetBets* which focuses on penny stocks, crowd sentiment is studied to be identified either as reliable signals or mere noise amplification during periods of heightened market attention. Their findings suggest that predictive performance varies across contexts and that the presence of highly active users and coordinated behaviour can influence aggregate outcomes [4]. While this work highlights heterogeneity in user behaviour, its analysis remains focused on the collection of sentiment dynamics rather than on identifying persistently informative individuals.

Reddit has emerged as an informative platform particularly for cryptocurrency analysis due to the longer character limit for posts and availability of data. The price of cryptocurrency is largely dependent on public opinion as it is not backed by governments or a company value, which explains the strong relationship between online conversations and the asset. Wooley demonstrated that sentiment extracted from Reddit discussions correlates with cryptocurrency price movements, with certain subreddits acting as early indicators of market trends, especially for Bitcoin and Ethereum [3]. Camou extends this line of work by incorporating Reddit-specific activity and engagement features, showing improvements in predictive performance for selected crypto-assets [2]. In both studies, predictive signals are derived primarily at the level of subreddits or the platform as a whole, rather than at the

level of individual users. Wooley further suggests that predictive information may originate from only a subset of community participants [3], motivating the development of models capable of identifying and characterising users whose contributions are consistently informative.

A prerequisite for any individual-level analysis, however, is the ability to reliably identify and extract directional forecasts from informal Reddit discourse, a substantially harder problem than sentiment classification on professional financial text. Existing work applying LLMs to financial language has operated primarily on structured sources such as professional news headlines [7], and financial news articles [8]. Social media text presents distinct challenges that this literature does not address. Reddit posts are colloquial, hedged, and embedded in conversational context, with directional claims that are implicit rather than explicit, temporally vague, and interspersed with speculation, humour, and noise. Extracting a structured forecast from such text requires a model capable of pragmatic inference beyond surface-level sentiment classification, and no existing work has validated such a pipeline specifically for cryptocurrency forecasting on Reddit. This methodological gap is a necessary precondition for the downstream user-level analysis pursued in this work.

### **2.1.2 Graph-Based Social Network Analysis**

The classical social network analysis literature commonly conceptualises social systems as graphs in which nodes represent individuals and edges encode social relationships or interactions. Within this framework, networks are characterised using structural properties such as degree distributions, clustering coefficients, path lengths, and community structure [5], which form the basis of many descriptive analyses of the collective patterns of interaction found in social systems.

Homophily has been shown to be a pervasive feature of social networks and a key network metric in understanding observed patterns of connectivity [9], (Equation 2.2). Information diffusion provides a concrete illustration of how graph structure encodes semantically meaningful signals independently of textual content. Credible and non-credible information tend to propagate through networks via structurally distinct patterns, producing diffusion trees that differ in depth, branching factor, and temporal spread, such that veracity can be inferred from the shape of the cascade alone [10]. This demonstrates that the relational structure of how content travels through a network, rather than what the content says, can itself carry information about its reliability, motivating the use of structural features as signals in content-identifying tasks.

More recently, graph representation learning and graph neural networks have been applied directly to social network analysis tasks. Graph representation learning methods map nodes to low-dimensional feature vectors that preserve structural information and can be used for downstream tasks such as classification and link prediction [11]. Graph neural networks extend this by defining neural archi-

tructures that aggregate information from neighbouring nodes, allowing representations to incorporate both structural context and node attributes [12], with semi-supervised variants enabling inference over unlabelled nodes from a limited labelled set [13]. Surveys of GNN applications to social networks document strong performance across tasks including community detection, influence estimation, and node classification when sufficient structural and attribute information is available [14, 15]. In social network settings specifically, node features are typically derived from observable behavioural signals such as posting frequency, content characteristics and graph structure, which provides complementary relational context that improves over feature-only approaches [15].

Taken together, the existing literature establishes that Reddit sentiment carries predictive information about cryptocurrency price movements, and that graph-based machine learning provides effective tools for node-level inference in social networks. However, three gaps remain unaddressed. First, no principled method exists for reliably extracting and evaluating individual-level forecasts from informal Reddit discourses. Second, Reddit-based cryptocurrency forecasting research derives predictive signals by aggregating contributions across entire user populations, treating individual heterogeneity as noise rather than as a source of structure, and no existing work has developed a scalable framework for identifying persistently informative individuals. Third, while graph machine learning has been applied to social networks for tasks such as community detection and link prediction, it has not treated individual forecasting accuracy as a learnable node-level attribute. The present work addresses each of these gaps in turn, and they motivate directly the three research questions introduced in Chapter 3.

## 2.2 Background

In order to encapsulate the interaction relationships of users and potential predictive structural signals, graphs is a key data structure used in this research. Understanding it and the common graph metrics will aid interpretation of patterns found in the network. Machine-learning (ML) model techniques will also aid understanding the classification tool in this research.

### 2.2.1 Graphs

Graph structures provide a powerful method of representation of complex systems by capturing both entities and the interactions between them. This data structure enables the extraction and analysis of relational information using a wide range of analytical techniques, including modern machine-learning methods [11]. Formally, a graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  consists of a set of nodes  $\mathcal{V}$ , representing objects or entities, and a set of edges  $\mathcal{E}$ , representing interactions between pairs of nodes. An edge connecting node  $u \in \mathcal{V}$  to node  $v \in \mathcal{V}$  is denoted as  $(u, v) \in \mathcal{E}$  [16].

Different graph types can be used to represent the characteristics of real-world networks with greater fidelity. A common and useful type is the weighted graph, a graph in which each edge is assigned a

number, called its edge-weight, which is useful in representing relationships with different importance levels. It is defined as  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$  where  $\mathbf{W} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$  is the weighted adjacency matrix, with entry  $W_{ij}$  giving the edge weight between nodes  $i$  and  $j$  (and  $W_{ij} = 0$  if no edge exists).

## Graphs Representation

Structural and relational arrangements of entities can be captured differently in various representations, each providing unique benefits tailored to the application. These representations are often expressed in matrix form, which facilitates their computational analysis.

**Adjacency Matrix**,  $\mathbf{A}$  is the foundational representation of a graph, encoding its local connectivity structure [5]. In an unweighted graph,  $\mathbf{A}_{i,j}$  is 1 if a connection between nodes  $i$  and  $j$  is present. However, in a weighted graph,  $\mathbf{A}_{i,j}$  will be equal to the corresponding edge-weight.

**Degree Matrix**,  $\mathbf{D}$  is a diagonal matrix with diagonal entries  $d_i = \sum_j \mathbf{A}_{ij}$ , representing the degree of node  $i$ . This representation captures node-level connectivity patterns and is frequently employed as a proxy for node importance or centrality in graph-based analysis.

## Graphs Features

While the structural properties of a graph encode relational information, additional inference can be extracted from feature representations that describe the attributes of nodes, edges, or entire graphs.

**Node Feature Matrix** is denoted by  $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times F}$  where each row  $\mathbf{x}_i$  represents the  $F$ -dimensional feature vector associated with node  $i$ . Node features provide semantic information that cannot be inferred from graph structure alone, enabling models to distinguish nodes with similar connectivity but different intrinsic properties. Incorporating node features is essential in improving machine-learning performances, particularly for social networks [17].

**Edge Features** can be represented by a function  $\phi : \mathcal{E} \rightarrow \mathbb{R}^K$ , where each edge is associated with a  $K$ -dimensional feature vector. These features are particularly beneficial when relationships are complex and multi-dimensional [18]. An example of an edge feature is the edge weight.

**Structural Features** are derived from the topology of the graph and encode information about a node's position and role within the network. Examples include degree, clustering coefficient (the density of connections among a node's neighbours), and path-based measures such as betweenness and closeness centrality (Section 2.2.2). Unlike node features, which are externally assigned attributes, structural features are computed entirely from the graph's connectivity pattern and carry no semantic information beyond relational position.

Graph learning methods typically combine structural information, encoded by the adjacency matrices, with feature information that captures node or edge attributes.

### 2.2.2 Centrality Measures

Centrality measures constitute a canonical class of structural features, quantifying a node’s positional importance within the graph. They are particularly informative for social network analysis [5], as they capture how influence, information access, and brokerage roles are distributed across users. The following measures are analysed in Section 5.1.1 and are used as node-level structural features in Section 5.2.2.

1. **Betweenness Centrality** measures the extent to which a node lies on the shortest paths between other nodes. High betweenness centrality identifies users who act as intermediaries or brokers between communities. (Equation in Appendix A.1)
2. **Closeness Centrality** quantifies how quickly a node can reach all others in the network, mathematically calculated via  $C(v) = \frac{n-1}{\sum_{u \neq v} d(v,u)}$  where  $d(v,u)$  is the shortest path distance between nodes  $v$  and  $u$ , and  $n$  is the number of nodes [19]. A user with high closeness centrality can disseminate or receive information more rapidly across the network.
3. **Eigenvector Centrality** assigns importance recursively, such that a node is influential if it is connected to other influential nodes. It is defined as the leading eigenvector of the adjacency matrix  $\mathbf{A}$  ( $\mathbf{Ax} = \lambda_{\max}\mathbf{x}$ ) where  $x_v$ , the  $v$ -th component of the principal eigenvector, gives the eigenvector centrality of node  $v$  [20].
4. **PageRank** is the stationary distribution of a random walk on the graph with a damping factor that introduces a uniform teleportation probability. This guarantees a unique solution and resolves the convergence issues that eigenvector centrality exhibits on graphs with disconnected components or sink nodes. (Equation in Appendix A.1)

### 2.2.3 Modularity and Homophily

Modularity and homophily are two complementary statistics for characterising the community structure of a network, capturing both the structural cohesion of communities and the degree to which that structure reflects underlying node similarity.

**Modularity** measures the quality of a specific network division into clusters, making it a suitable variable to maximise in a community detection algorithm. It quantifies the strength of division by comparing the number of edges within clusters as compared to the expected number in a random network [21]. A high modularity indicates a strong community structure. The modularity  $Q$  is defined as:

$$Q = \frac{1}{2m} \sum_{i,j} \left( A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j) \quad (2.1)$$

where  $A_{ij}$  is the adjacency matrix entry for nodes  $i$  and  $j$ ;  $k_i$  and  $k_j$  are the degrees of nodes  $i$  and  $j$ ;  $m$  is the total number of edges in the network;  $c_i$  and  $c_j$  represent the community assignments of

nodes  $i$  and  $j$ ; and  $\delta(c_i, c_j)$  is the Kronecker delta function.

**Homophily** in a graph is defined as the tendency of nodes to connect with other nodes that share the same attributes. This can be quantified as:

$$h = \frac{\sum_{(i,j) \in E} w_{ij} \mathbb{I}(y_i = y_j)}{\sum_{(i,j) \in E} w_{ij}}, \quad (2.2)$$

where  $w_{ij}$  denotes the edge weight and  $\mathbb{I}(\cdot)$  is the indicator function [9].

At the node level, homophily for node  $i$  is defined as:

$$h_i = \frac{\sum_{j \in \mathcal{N}(i)} w_{ij} \mathbb{I}(y_i = y_j)}{\sum_{j \in \mathcal{N}(i)} w_{ij}}, \quad (2.3)$$

with  $\mathcal{N}(i)$  denoting the neighbourhood of node  $i$ . Homophilic graphs tend to have a higher modularity, since similar nodes connect to one another more frequently, creating stronger clusters.

#### 2.2.4 Learning on Graphs

Graph-based machine-learning exploits the dependencies induced by graph structure directly in its predictions, in contrast to traditional settings where data instances are assumed to be independent and identically distributed (i.i.d.) [11]. The primary task considered in this work is node classification, where the task is to predict a label for each node by leveraging both node features and neighbourhood structure. This is formulated as a semi-supervised problem, where labels are available for only a subset of nodes, and graph topology acts as an inductive bias that promotes label smoothness across connected nodes [13]. The effectiveness of this formulation depends on the degree of homophily in the graph (the tendency for connected nodes to share labels, defined formally in Equation 2.2) though architectures designed for heterophilic settings, where connected nodes differ in label, are also considered [22].

#### Message-Passing Framework

The central computational primitive in graph-based learning is message-passing. At each layer  $\ell$ , every node  $u$  updates its representation by aggregating information from its neighbours  $\mathcal{N}(u)$ .

$$h_u^{(\ell+1)} = \sigma \left( W^{(\ell)} \cdot \text{AGGREGATE} \left( \left\{ h_u^{(\ell)} : u \in \mathcal{N}(u) \right\} \right) + b^{(\ell)} \right), \quad (2.4)$$

where  $h_u^{(0)} = x_u$  is the initial node feature vector,  $W^{(\ell)} \in \mathbb{R}^{d_{\ell+1} \times d_\ell}$  and  $b^{(\ell)} \in \mathbb{R}^{d_{\ell+1}}$  are learnable parameters, and  $\sigma$  is a non-linear activation. After  $L$  layers,  $h_u^{(L)}$  encodes information from the  $L$ -hop neighbourhood of  $u$ , combining structural context with node-level attributes. The choice of AGGREGATE function gives rise to the specific architectures evaluated in Chapter 6.

### 3 Problem Formulation

This project investigates how user interactions and network position within *r/CryptoMarkets* relate to predictive capability in cryptocurrency discussions. The three gaps identified in Section 2.1 motivate the following research questions:

1. How effectively can language models extract and evaluate directional forecasts from informal Reddit discourse?
2. Do the social network interaction patterns and behavioural characteristics of a user indicate their ability to predict cryptocurrency market movements?
3. Which structural and behavioural features enable graph-based models to identify users with high predictive potential, including those with limited historical activity?

**Hypothesis** Building on evidence that predictive signals originate from a subset of participants [3] and that network position is informative about node-level attributes [5], it is hypothesised that high-predictive users are structurally identifiable through their interaction patterns, for instance via heterophilic connectivity or distinctive centrality.

**Learning Task Formulation** Let  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$  be the weighted social interaction graph over users  $\mathcal{V}$ , with  $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times F}$  the node feature matrix of behavioural, textual, and structural attributes. For each user  $u$ , a language model extracts directional forecasts from their posts and comments, the prediction is evaluated against realised market returns to produce a composite score. Users are assigned binary *High/Low* labels  $\ell_u \in \{0, 1\}$  based on whether this score exceeds a threshold. Labels are available only for  $\mathcal{V}_L \subset \mathcal{V}$ , the subset of users with sufficient forecast history, the remaining  $\mathcal{V}_U = \mathcal{V} \setminus \mathcal{V}_L$  are unlabelled.

The core task is a semi-supervised node classification, where graph topology propagates information from labelled to unlabelled nodes. Given  $(\mathcal{G}, \mathbf{X}, \{\ell_u\}_{u \in \mathcal{V}_L})$ , learn a classifier that generalises to unlabelled nodes  $\mathcal{V}_U$ , with graph topology acting as a structural prior via the message-passing framework in Equation 2.4. RQ1 addresses the reliability of the extraction pipeline, RQ2 examines whether labels associate with observable interaction patterns, and RQ3 identifies which features of  $\mathbf{X}$  and  $\mathcal{G}$  are most discriminative.

## 4 Data Collection and User Labelling

This chapter describes the pipeline for transforming raw Reddit discourse into a labelled social interaction graph suitable for graph-based machine learning. The process spans five stages: data collection and cleaning, forecast extraction, LLM-based validation, market return evaluation, and user scoring. Figure 4.1 provides an overview of this pipeline, detailing each stage in the sections that follow.

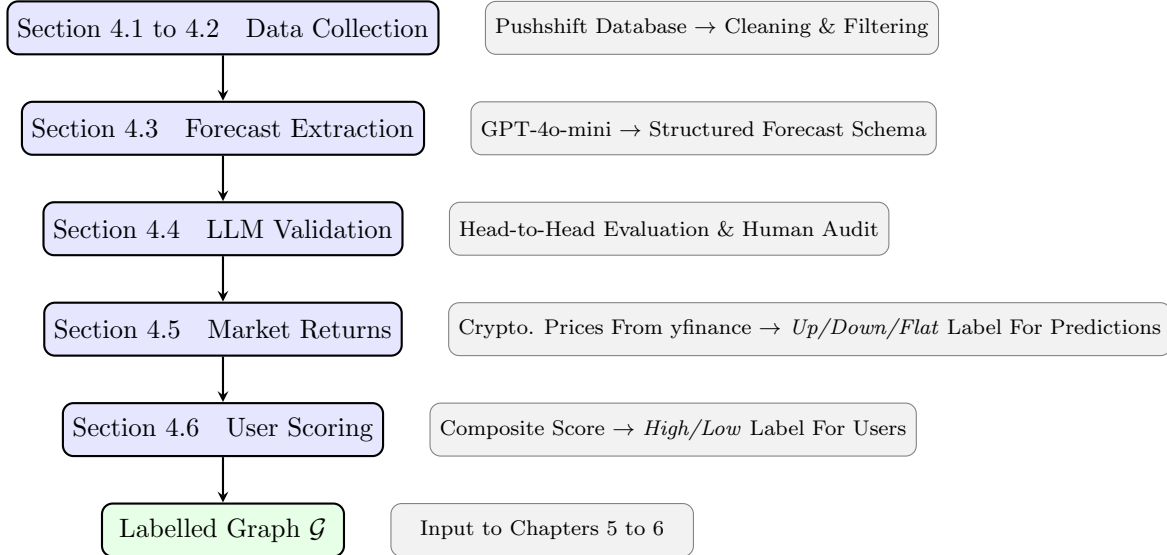


Figure 4.1: Overview of the data collection and user labelling pipeline presented in Chapter 4. Each stage is detailed in the section indicated. The output is the labelled social interaction graph  $\mathcal{G}$ .

### 4.1 Reddit Dataset

Reddit is a social media platform where users can create public posts and comments on the posts, as well as on other comments. Users can demonstrate their approval or disapproval of posts and comments through upvotes or downvotes, which when collected can earn themselves “*karma points*”. This encourages users to post good content, make useful comments and provide relevant feedback [23]. All posts on Reddit are submitted under a “*subreddit*”, a forum where a community is focused on a specified topic. This makes it easy to obtain subject-specific datasets.

#### **r/CryptoMarkets Subreddit**

This study focuses on the subreddit *r/CryptoMarkets*, which aims to keep investors updated on the ever-changing world of crypto trading and investing. Compared to other popular cryptocurrency subreddits such as *r/Bitcoin* and *r/ethereum*, *r/CryptoMarkets* focuses on a variety of cryptocurrency assets, and the whole cryptocurrency market as a whole. With strict rules such as *Rule #5 - No Low Quality Content* and moderators enforcing them, the subreddit has a vast collection of news, updates and financial analysis posted regularly. *r/CryptoMarkets* has 1,900,000 members who are a part of the subreddit. It sees 134,000 visitors and 3,800 contributions every week<sup>1</sup>, giving sufficient data to find

<sup>1</sup>Data was recorded from Reddit, according to the significant figures shown.

a variety of analysis qualities, and user-user interactions to build an informative graph of the network between the users on the subreddit.

## 4.2 Data Processing

### 4.2.1 Data Extraction

Submission and comment data from *r/CryptoMarkets* is collected from Pushshift Reddit database [24] which provides a regularly updated archive of all posts and comments on Reddit. For this project, the data extracted is from June 2024 to June 2025. Each month consists of a data set that is large enough to support analysis. For instance, the data set for June 2025 consists of 40,421,082 posts and 743,406 unique subreddits.

The dataset contains the core metadata required for social network analysis, such as subreddit identifiers, post titles, comment bodies, edit status, engagement scores, and author usernames. Author usernames serve solely as node identifiers for constructing interaction edges and are not disclosed or analysed individually. No private identifying information such as email addresses, IP addresses, or account credentials is collected, and all data is drawn from publicly available Reddit content via the Pushshift archive. Ethical considerations are discussed further in Section 8.3.2. Engagement scores refer to the net Reddit vote count of a post or comment, computed as the number of upvotes minus downvotes, which serves as a community-level signal of content quality and visibility.

### 4.2.2 Network Construction

The data was streamed and stored in Apache Parquet format using the PyArrow library. Parquet’s columnar storage enables efficient compression and scalable processing, making it well suited for large-scale Reddit datasets.

To best encapsulate the user-user relationships on the “*r/CryptoMarkets*” subreddit, the data is then stored in a graph. An undirected weighted graph is used to model the user interaction network, where nodes represent individual Reddit users, and edges represent interactions formed when one user comments on another user’s posts. Edge weights capture the frequency of interactions from either node connected, such that repeated engagement between the same pair of users results in a stronger weighted connection.

NetworkX was used to construct and analyse graph structures. NetworkX is a Python package for the creation, manipulation, and study of the structure, dynamics, and functions of complex networks [25]. NetworkX allows us to initialise a graph, add nodes and edges, compute centrality metrics, and visualize the graph structure. In-depth graph analysis is found in Chapter 5.

For visual inspection and exploration, the final network was exported in GEXF and CSV formats and analysed using Gephi. Gephi was selected for its interactive layouts, real-time filtering, and ability to

handle large attributed networks, enabling effective examination.

### 4.2.3 Data Cleaning

Prior to analysis, the Reddit data was subjected to a comprehensive cleaning process to reduce noise and remove content that could bias network structure and centrality measures. Social media data frequently contains many signals, a majority of which can heavily distort the signal from the interaction patterns of genuine users. The actions taken to mitigate noise is listed in Table 4.1.

Table 4.1: Methodological filtering criteria and thresholds

| Filter Category     | Threshold                                     | Methodological Justification                                                                                           |
|---------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Min. Title Length   | 10 characters                                 | Ensures submissions contain enough information to facilitate sentiment analysis.                                       |
| Min. Body Length    | 20 characters                                 | Same as above.                                                                                                         |
| Min. Engagement     | Score <sup>†</sup> $\geq 10$ ; Comms $\geq 1$ | Filters out ignored or isolated content.                                                                               |
| Max. Text Length    | 10,000 characters                             | Filters out spam and eliminates distortion of length-based metrics from automated sources.                             |
| Min. Network Degree | $k \geq 3$                                    | Ensures the graph represents a highly interactive social network.                                                      |
| Max. Edge Weight    | $w \leq 100$                                  | Filters users with bot-like relationships.                                                                             |
| Degree Centrality   | $C_D \leq 0.5$                                | Filters star topologies commonly exhibited by bots, which is unlike the tree structures found with genuine users [26]. |
| Account Limits      | Max 1000C / 100P                              | Excludes high-frequency automated broadcasting accounts.                                                               |
| Text Similarity     | 0.9 Title / 0.85 Body                         | Uses Jaccard similarity to exclude reposted material, preventing inflated user influences.                             |
| Min. Edge Weight    | $w_{min} = 3$                                 | Ensures a minimum level of interaction strength.                                                                       |

<sup>†</sup> Engagement score = net vote count (upvotes – downvotes) assigned by the Reddit community.

Collectively, these steps ensures that the final dataset primarily reflected meaningful and authentic user discussion, and a signal-to-noise ratio high enough for machine learning models to discern any pattern available. This resulted in an overall of 878 posts and 124,292 comments from 2,617 users.

## 4.3 Forecast Extraction

To evaluate user predictive performance, qualitative social media discourse must be transformed into quantitative structured signals. This section details the forecast taxonomy, the extraction schema, and a comparative validation of labelling methodologies.

### 4.3.1 Forecast Taxonomy and Filtering

A forecast is any submission or comment that asserts or implies a future directional movement, focusing particularly on cryptocurrency prices in this project. This includes explicit predictions (*e.g.* “*BTC will rally next week*”) and implicit directional views expressed through trading intent (*e.g.* “*I am accumulating here*” as a *bullish signal*). General news dissemination, meta commentary and scams are classified as *non-forecasts* because they do not provide an evaluable directional claim.

*Non-forecasts* and *neutral* posts are excluded from the scoring pipeline to reduce label noise and avoid attributing predictive skill to content that cannot be scored against realised market movements. However, these posts are retained during network construction, since removing them would discard

genuine user interactions and distort connectivity patterns. Preserving these structural signals is important for downstream models that learn from user behaviour and position within the network.

### 4.3.2 Signal Representation

To enable systematic market alignment scoring, each forecast is mapped to a fixed structured schema, detailed in Table 4.2.

Table 4.2: Extracted forecast fields using LLM

| Field                   | Description                                                                                                                                      | Examples / Range                                         |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Direction</b>        | Implied direction of future price movement.                                                                                                      | Bullish, Bearish                                         |
| <b>Assets</b>           | Referenced cryptocurrencies mapped to canonical tickers. <i>Allusions to the broader market were assigned a market-wide identifier: Bitcoin.</i> | BTC, ETH, SOL, etc.                                      |
| <b>Time Horizon</b>     | Coarse temporal classification of the forecast.                                                                                                  | Short-term, Mid-term, Long-term, Cyclical, Event-driven  |
| <b>Timeframe Detail</b> | Minutiae temporal descriptors that refine the broad <b>Time Horizon</b> category.                                                                | End of month, post-CPI release, after ETF approval, etc. |
| <b>Confidence Score</b> | Inferred author confidence in the forecast.                                                                                                      | 0 to 100                                                 |

This structure allows the labelled outputs to be passed directly into the market-alignment scoring stage, while preserving the integral temporal qualifiers for interpreting and evaluating each forecast.

### 4.3.3 Labelling Methods

Three methodologies for the labelling of the forecasts were evaluated for scalability and fidelity:

- **Manual Labelling:** Served as the high-fidelity reference standard but not scalable for large corpora.
- **Heuristic Labelling:** A rule-based baseline using fixed lexicons. This approach proved insufficient for informal Reddit discourse (full specification and explanation in Appendix A.2).
- **LLM Labelling (GPT-4o-mini & GPT-4o):** Adopted as the primary extraction method, which is justified in Section 4.4. Reliability was enhanced via deterministic decoding, fixed prompts, strict JSON schema enforcement, and automated validation with retry logic.

## 4.4 LLM-based Forecast Validation

To address the limitations of heuristic extraction and the scalability constraints of manual labelling, LLM-based labelling was adopted as the primary extraction method. GPT-4o and GPT-4o-mini were selected due to the availability of departmental funding. Both models were released by OpenAI in 2024, with GPT-4o serving as the high-capability general model and GPT-4o-mini providing a more computationally efficient and economical alternative. To ensure a controlled comparison, both models were evaluated using an identical structured extraction prompt.

### 4.4.1 Head-to-Head Model Evaluation

To evaluate the model against the ground truth, a dataset of 278 posts was extracted from *r/CryptoMarkets* (June 2025) and manually annotated. Unlike a random shuffle, this stratified temporal subset pre-

serves the natural distribution of sentiment volatility and linguistic nuances inherent in cryptocurrency discussions, in proportions consistent with the longitudinal data. This approach mitigates biases and tests the model’s ability to generalise across the Reddit colloquialisms. Model validation was measured using Macro-F1 on the extracted **Direction** from the labelled posts and comments, where  $\text{Direction} \in \{\text{Bullish}, \text{Bearish}\}$ .

The F1-score is defined as:

$$F1 = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}}, \text{ where } \text{Precision} = \frac{TP}{TP + FP} \text{ and } \text{Recall} = \frac{TP}{TP + FN} \quad (4.1)$$

TP is True Positive, FP is False Positive and FN is False Negative.

Macro-F1 is defined as such:

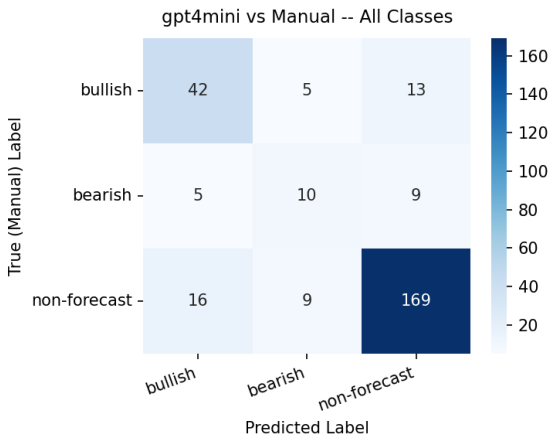
$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C F1_c, \quad (4.2)$$

where C is the number of classes. The F1-score accounts for both FPs and FNs well as it takes the harmonic mean of both precision and recall [27]. Precision, recall and accuracy can cause inflated performances from a majority class bias, which can lead to costly trading decisions. Macro-averaging mitigates this bias further through the equal weighting across the classes.

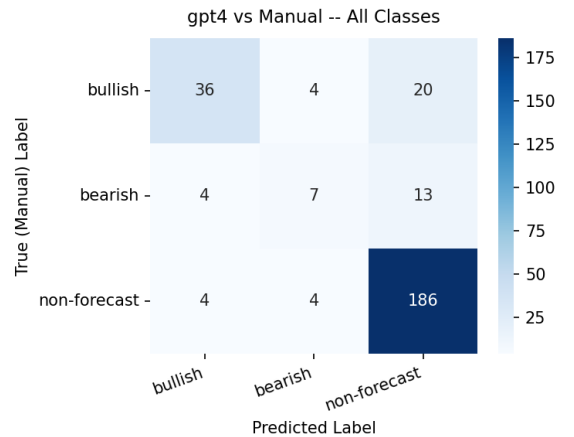
Table 4.3: Forecast-only Macro-F1 Comparison

| Model       | Macro-F1      |
|-------------|---------------|
| GPT-4o      | 0.7682        |
| GPT-4o-mini | <b>0.7801</b> |

The scores are recorded in Table 4.3. GPT-4o-mini achieves a marginally higher forecast-only macro-F1 score, suggesting that it is more effective at correctly identifying directional forecasts while maintaining robustness across both bullish and bearish labels.



(a) GPT-4o-mini confusion matrix



(b) GPT-4o confusion matrix

Figure 4.2: Confusion matrices for GPT-4o-mini (left) and GPT-4o (right) against manual ground truth across bullish, bearish, and non-forecast classes. GPT-4o-mini achieves higher macro-F1 (0.7801 vs 0.7682) with stronger performance on forecast-relevant classes.

The confusion matrices in Figure 4.2 further support this result. GPT-4o-mini shows a stronger diagonal for the forecast classes, with fewer bullish instances misclassified as non-forecasts and improved identification of bearish predictions. Although GPT-4o performs marginally better on the non-forecast class, GPT-4o-mini achieves superior discrimination on the forecast directions, which are the focus of this evaluation.

To better understand model behaviour beyond aggregate metrics, a structured disagreement analysis was also conducted. Posts were marked as contested where there was a disagreement between GPT-4o predictions, GPT-4o-mini predictions, and manually assigned ground-truth labels. These contested cases primarily arose in borderline forecasts with a weak direction indication, polysemous signals and combinations of discussion and prediction. This analysis highlights that disagreement is concentrated in inherently ambiguous content rather than clear forecasting statements, reinforcing the reliability of both models on well-defined signals. This result is not unexpected, as LLMs tend to exhibit greater inconsistency when differentiating vague language.

Based on the superior macro-F1 performance and comparable qualitative behaviour, GPT-4o-mini was selected as the primary model to label the full dataset of posts and comments. In addition to its improved predictive performance, GPT-4o-mini offers significantly lower computational cost, making it more suitable for large-scale labelling.

#### 4.4.2 Human Audit and Validation

To further validate model reliability, a human audit was conducted on a stratified random sample of the posts labelled by GPT-4o-mini. The audit process involved manually assigning ground-truth labels on 50 random posts out of the labelled extracted from the year-long 878 canonical posts, and comparing them against model predictions.

Table 4.4: Human audit vs GPT-4o-mini labelling performance

| Metric         | Value                                   |
|----------------|-----------------------------------------|
| Accuracy       | 0.8400                                  |
| Macro-F1 Score | 0.8505                                  |
| Cohen’s Kappa  | 0.7587 ( <i>substantial agreement</i> ) |

Note: A Cohen’s kappa between 0.61 and 0.80 indicate substantial agreement between annotators.

Evaluation metrics included accuracy, macro-averaged F1-score, and Cohen’s kappa, providing a complementary assessment of model performance. Accuracy captures overall correctness, macro-F1 was defined in Equation 4.2, and Cohen’s kappa measures agreement beyond chance. The consistency of high values across all three metrics indicates both strong classification performance and substantial agreement with human annotations.

The confusion matrix in Figure 4.3 further supports the strong agreement between human annotations and model predictions. The model achieves perfect classification for bearish instances, with no

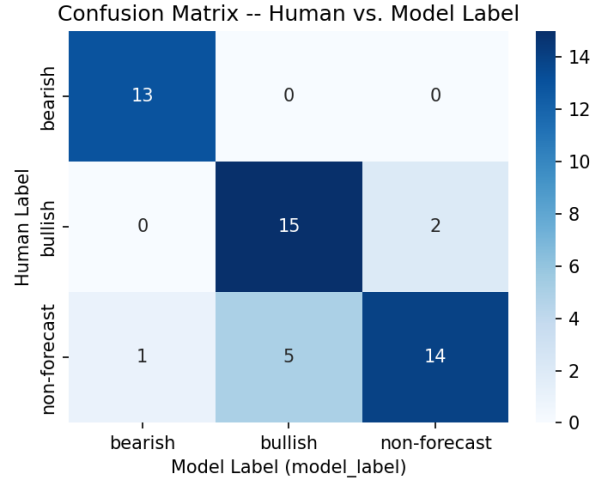


Figure 4.3: Confusion matrix of GPT-4o-mini predictions against human annotations on 50 posts. Bearish instances are classified perfectly (13/13); primary errors occur in the non-forecast class, where 6/20 instances are over-interpreted as forecasts.

observed misclassification, suggesting consistent identification of bearish sentiment. Similarly, bullish predictions are classified with a high accuracy, with only minor confusion into the non-forecast class, indicating occasional ambiguity in distinguishing implicit bullish signals from non-predictive content. Most disagreement arises within the non-forecast class, where 6 out of 20 instances are misclassified as forecast-related. This suggests that the primary source of error lies in the model’s tendency to over-interpret ambiguous language as actionable forecasts. The results demonstrate strong directional classification performance, with residual errors concentrated in the boundary between forecast and non-forecast content rather than between bullish and bearish classes.

Overall, approximately 15–20% of forecasts are likely to be misclassified, with errors concentrated at the forecast and non-forecast boundary rather than between bullish and bearish classes. These boundary errors are partially averaged out at the user level by the aggregation in Section 4.6, but the residual sensitivity of label assignments to this error rate is discussed in Section 8.3.1. Collectively, the quantitative evaluation, disagreement analysis, and human audit support the use of GPT-4o-mini as the primary extraction model for the full dataset.

#### 4.4.3 Analysis of Extracted Forecasts

Of all the 5 fields extracted with the LLM, the confidence score is the most unique. High-confidence predictions tend to be characterised by specific asset references, explicit directional claims, and supporting reasoning, whereas low-confidence predictions are typically vague, often using hypotheticals or vocabulary indicating a lack of certainty. Representative examples are provided in Appendix A.3.

Figure 4.4 provides an overview of the composition of labelled posts and comments.

The direction distribution is heavily skewed towards non-forecast content (58%), indicating that the majority of discussion does not contain actionable predictions. Among forecast-related content, bullish

#### Forecast Composition: Labelled Posts / Comments Analysis (n = 125,170)

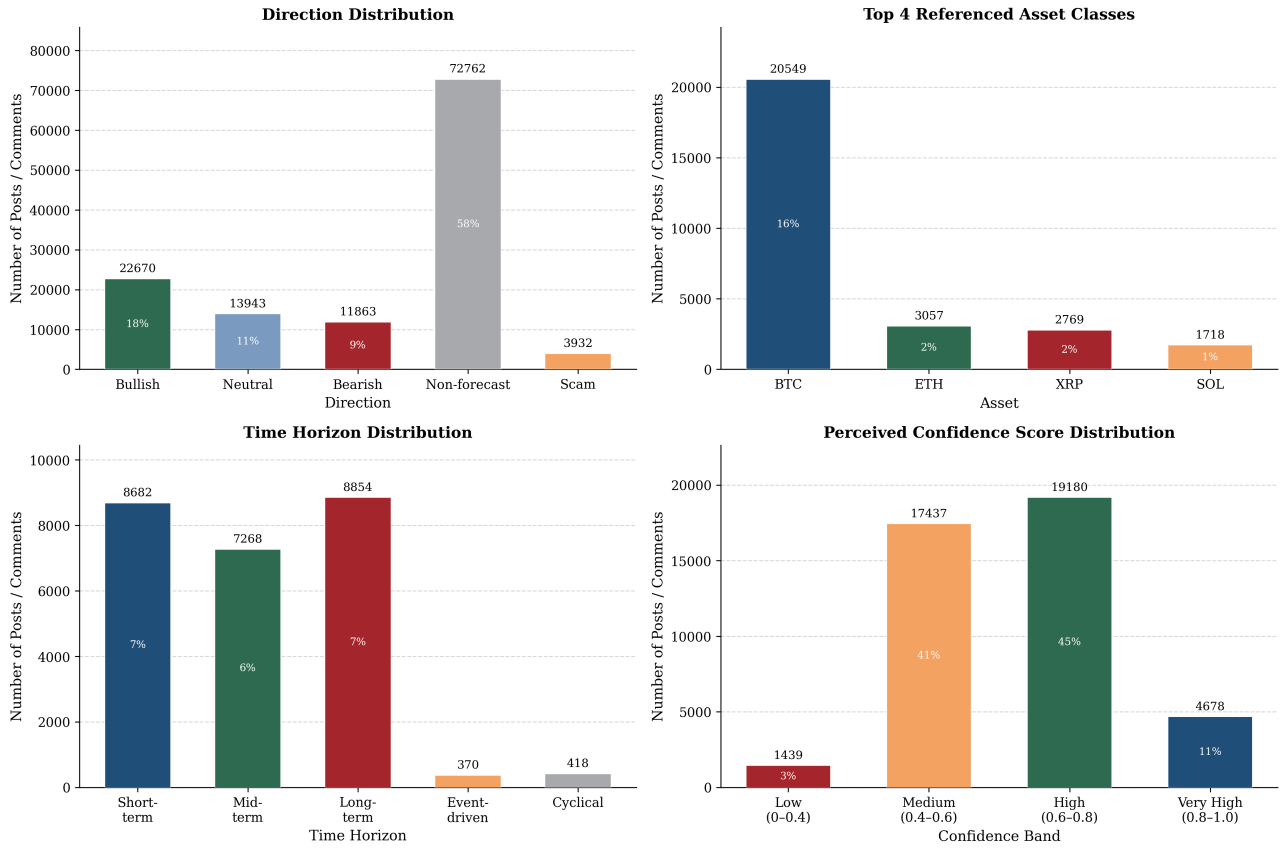


Figure 4.4: Composition of 125,170 labelled posts and comments across four dimensions: direction (bullish 18%, bearish 9%); asset references (BTC dominates at 16%); time horizon (broadly distributed across short, mid, and long term); and perceived confidence (concentrated in medium and high bands)

signals (18%) are more prevalent than bearish signals (9%), suggesting an overall optimistic bias within the community, which one might expect from a community focusing on cryptocurrency. Neutral (11%) and scam-related content (3%) form a smaller proportion of the dataset.

Asset references are highly concentrated, with Bitcoin (BTC) dominating discussions (16% of all posts and comments), while other major assets such as Ethereum (ETH), XRP, and Solana (SOL) appear far less frequently (1–2% each). This ascertains that BTC and the cryptocurrency market as a whole are the primary focus of discourse on *r/CryptoMarkets*.

The distribution of time horizons shows a relatively balanced spread across short-term (7%), mid-term (6%), and long-term (7%) forecasts, with a slight emphasis on longer horizons. In contrast, event-driven and cyclical predictions are rare (both below 1%), indicating that most forecasts are framed in general temporal terms rather than tied to specific events or recurring patterns.

Finally, perceived confidence levels are strongly skewed towards medium (41%) and high (45%) bands, with relatively few low-confidence predictions (3%). This suggests that users tend to avoid expressing highly uncertain views, while still rarely committing to extreme certainty (very high confidence: 11%).

## 4.5 Calculation of Market Returns

To evaluate whether a Reddit post or comment comprises of a correct directional forecast, the extracted fields are used to ascertain the realised market return over the implied horizon.

### 4.5.1 Price Data Retrieval

In order to retrieve the price of the cryptocurrency assets from yfinance, an open-source Python library for retrieving market data from Yahoo Finance [28], assets are normalised and converted to a yfinance ticker via an alias map. Prediction horizons are converted to hours via a base mapping through the `Time Horizon` field, and may be overridden by the `Timeframe Detail` field. The final horizon is clamped to a valid range to prevent extreme horizons. The temporal resolution of the price series is selected dynamically based on the prediction horizon. For short-term forecasts (i.e. horizons  $\leq 14$  days), hourly data is used to capture fine-grained intra-day market dynamics. For longer-term forecasts, daily data is adopted to mitigate high-frequency noise and provide more stable return estimates. All timestamps are standardised to Coordinated Universal Time (UTC), and missing observations are handled using an *as-of* lookup, whereby prices are forward-filled to the most recent available observation at or prior to the target timestamp. Historical adjusted close prices are fetched from yfinance to compute both the forward return and the historical benchmark distribution.

### 4.5.2 Simple Returns and Abnormal Returns

Let  $t$  be the post creation time, and  $H$  the final prediction horizon in hours. Using the price series  $P(\cdot)$ , the realised simple return is

$$r_{t,H} = \frac{P(t+H) - P(t)}{P(t)}. \quad (4.3)$$

If either endpoint price is unavailable, the return is invalid for that post. This could occur if yfinance does not record price data for niche alt-coins.

To account for the impact of assets and time horizons on the volatility, a historical benchmark is computed from rolling forward returns over the same horizon using a lookback window (90 days). 90 days is a domain heuristic, as it is long enough to capture the volatility of the asset while short enough to remain representative of the current market regime. Denoting the benchmark mean and standard deviation as  $\mu_H$  and  $\sigma_H$  respectively, the abnormal return and its standardised form are

$$\text{AR}_{t,H} = r_{t,H} - \mu_H, \quad z_{t,H} = \frac{\text{AR}_{t,H}}{\sigma_H} \quad (\sigma_H > 0), \quad (4.4)$$

with  $z_{t,H} = 0$  if  $\sigma_H = 0$ . In addition, an approximate percentile rank is computed by interpolating between benchmark quantiles to contextualise the realised move with historical movements.

### 4.5.3 Market Movement and Prediction Classification

The realised movement is discretised into *Up/Down/Flat* using a hybrid threshold:

$$\tau = \max(\tau_0, 0.5 \sigma_H), \quad (4.5)$$

where  $\tau_0$  is an absolute neutral threshold of 0.25%. It is a practical floor that accounts for noise, avoiding mislabelling negligible price movement to be labelled as meaningful direction outcomes. If  $|r_{t,H}| < \tau$  the movement is *Flat*, otherwise it is *Up* if  $r_{t,H} > 0$  and *Down* if  $r_{t,H} < 0$ . This benchmark was a design heuristic, combining an absolute floor with volatility scaling. This prevents the decision rule from being too lenient in low-volatility periods and too strict in high-volatility periods.

Each post's predicted direction (Bullish/Bearish) is then mapped to *Up/Down* and compared against the classified realised movement to produce a binary label `prediction_correct`  $\in \{0, 1\}$ .

A stricter label `prediction_enhanced_correct` is also computed.

$$\text{prediction\_enhanced\_correct}_{t,H} = \begin{cases} 1, & \text{direction is bullish and } z_{t,H} > \gamma, \\ 1, & \text{direction is bearish and } z_{t,H} < -\gamma, \\ 0, & \text{otherwise,} \end{cases} \quad \gamma = 1.0. \quad (4.6)$$

This corresponds to one unit of standard deviation, which is a common measure that distinguishes between variations and significant deviations that are likely due to a specific trend or event.

### 4.5.4 Analysis of Predictions

Figure 4.5 presents the breakdown of correct and incorrect predictions grouped by forecast direction. The difference of `prediction_correct` and the more stringent `prediction_enhanced_correct` evaluation criteria can be observed. Under `prediction_correct`, bearish predictions achieve higher accuracy (52.7%) than bullish predictions (46.9%). However, when applying `prediction_enhanced_correct`, the accuracy drops substantially for both classes (28.6% for bullish and 35.6% for bearish). This reduction reflects the increased difficulty of identifying predictions that not only match direction but also capture meaningful market movements beyond typical volatility. Notably, the gap between bullish and bearish performance persists under both evaluation schemes, with bearish predictions consistently outperforming bullish ones. Enhanced bearish forecasts not only require significant downward price movements to be validated, but users who make such predictions also demonstrate greater accuracy in anticipating the timing of such market cycles. This highlights the volatility of the cryptocurrency market, as it experiences both bullish and bearish cycles, and the discrepancy between mere directional correctness and predictive strength. While a substantial proportion of forecasts align with

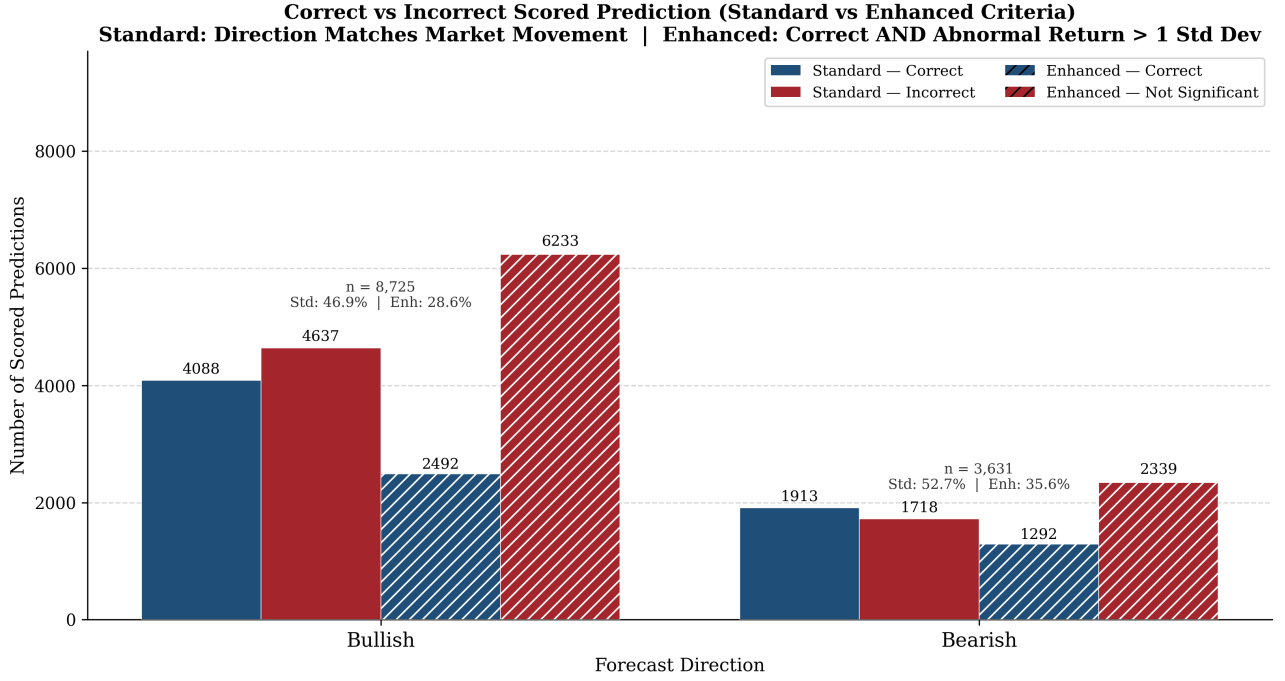


Figure 4.5: Prediction accuracy by forecast direction under standard (direction matches market) and enhanced (direction correct and abnormal return > 1 std dev) criteria. Bearish predictions outperform bullish under both criteria (52.7% vs 46.9% standard, 35.6% vs 28.6% enhanced), with qualifying forecasts dropping substantially under the stricter evaluation for both sentiments.

market direction, a much smaller subset correspond to significant abnormal returns, underscoring the importance of incorporating magnitude-aware evaluation metrics in assessing predictive quality.

## 4.6 Scoring Users

After each eligible post is evaluated against the market movement, users are scored by aggregating their prediction-level performance. The scoring pipeline implements confidence-weighted accuracy, abnormal-return-aware weighting, Bayesian shrinkage and final normalisation and tiering.

### 4.6.1 Prediction-Level Scores

Each prediction includes a perceived user confidence,  $c \in [0, 1]$  and a correctness indicator,  $y \in \{0, 1\}$ .

Three prediction-level scores are computed.

1. **Confidence-Weighted Score:**  $s^{(\text{conf})} = c \cdot y$ .
2. **Abnormal Return Score:**  $s^{(z)} = c \cdot y \cdot w(z)$ , where  $z$  is the standardised abnormal return, and factor  $w(z) = \text{clip}(1 + |z|, 0.1, 2.0)$  acts as a robustness measure, ensuring that extreme abnormal returns do not exert disproportionate influence on the final score.
3. **Enhanced Correct Score:**  $s^{(\text{enh})} = c \cdot y_{\text{enh}}$ , where  $y_{\text{enh}}$  is `prediction_enhanced_correct` (Equation 4.6).

The 3 scores are individually aggregated for each user  $u$ , to produce the user aggregated score,  $\bar{s}_u^{(m)}$ .

**User Aggregated Score.** For each user  $u$ , let  $n_u$  be the number of valid predictions.

$$\bar{s}_u^{(m)} = \frac{1}{n_u} \sum_{k=1}^{n_u} s_k^{(m)}, \quad m \in \{\text{conf}, \text{z}, \text{enh}\} \quad (4.7)$$

#### 4.6.2 Regularisation

To reduce variance in a data-rich project like this study, regularisation was implemented through bayesian shrinkage and normalisation.

Empirical Bayes shrinkage is used to reduce the instability of raw per-user scores  $\bar{s}_u^{(m)}$ , yielding:

$$\tilde{s}_u^{(m)} = \frac{n_u \bar{s}_u^{(m)} + \alpha_m \mu_0^{(m)}}{n_u + \alpha_m}, \quad (4.8)$$

where  $\alpha_m$  controls the shrinkage strength by acting as an equivalent number of pseudo-observations, anchoring the estimate toward the global prior mean  $\mu_0^{(m)}$  before sufficient user-level data has accumulated. At any  $n_u$  predictions, the effective weight placed on the user’s own observed score is  $n_u/(n_u + \alpha_m)$ , approaching 1 only as  $n_u$  grows large, and a larger  $\alpha$  corresponds to stronger shrinkage [29]. Two values are used:  $\alpha = 10$  for the confidence-weighted and enhanced-weighted metrics, and  $\alpha = 5$  for the z-score-weighted metric. The confidence-weighted and enhanced-weighted scores receive heavier shrinkage because each individual prediction provides relatively weak evidence, and hence potentially more noise. The z-score-weighted metric uses lighter shrinkage as a z-score above 1 is itself diagnostic of forecast quality, so user-level estimates can be trusted with fewer observations.

The Bayesian-adjusted scores  $\tilde{s}_u^{(m)}$  are min–max normalised to  $[0, 1]$  to make metrics comparable:

$$\hat{s}_u^{(m)} = \frac{\tilde{s}_u^{(m)} - \min_v \tilde{s}_v^{(m)}}{\max_v \tilde{s}_v^{(m)} - \min_v \tilde{s}_v^{(m)}}. \quad (4.9)$$

#### 4.6.3 Composite Score and User Tiers

A final composite score is computed as a weighted combination of the normalised metrics, defined as:

$$\text{composite}_u = 0.4 \hat{s}_u^{(\text{conf})} + 0.4 \hat{s}_u^{(\text{z})} + 0.2 \hat{s}_u^{(\text{enh})}. \quad (4.10)$$

The composite score weights (0.4,0.4,0.2) place equal emphasis on directional accuracy and confidence-weighted accuracy, as capturing such performance is essential to determining one’s predictive ability. The enhanced component receives a smaller weight as a secondary additional reward for strong correct calls. Users were then assigned *High* if their composite score is above 0.66 on the  $[0,1]$  scale, and will be assigned *Low* otherwise.

# 5 Network and Label Analysis

## 5.1 Social Interaction Graph

### 5.1.1 Network Statistics

The structural properties of the user interaction graph were analysed to characterise the overall organisation of the *r/CryptoMarkets* discussion network. Table 5.1 summarises the principal statistics, computed on  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$  with the following weighted adaptations. **Degree** counts distinct neighbours  $|\mathcal{N}(u)|$ , while **weighted degree** sums incident edge weights  $\sum_{v \in \mathcal{N}(u)} w_{uv}$ . **Clustering coefficient** uses the weighted generalisation of [30], replacing binary adjacency with the geometric mean of edge weights (Equation in Appendix A.1). The **small-world coefficient**,  $\sigma = (C/C_r)/(L/L_r)$ , compares clustering and path length against an equivalent random graph [31].

Table 5.1: Network statistics and homophily analysis

| Category           | Metric                              | Value  | Category          | Metric           | Value  |
|--------------------|-------------------------------------|--------|-------------------|------------------|--------|
| <b>Topology</b>    | Nodes ( $ V $ )                     | 2,617  | <b>Centrality</b> | Mean Closeness   | 0.29   |
|                    | Edges ( $ E $ )                     | 17,937 |                   | Mean Betweenness | 0.0009 |
|                    | Avg. Degree ( $\langle k \rangle$ ) | 12.89  | <b>Modularity</b> | $Q$              | 0.34   |
|                    | Density                             | 0.0052 | <b>Homophily</b>  | Overall (Node)   | 0.87   |
| <b>Small-World</b> | Avg. Path Length ( $L$ )            | 2.97   |                   | High-pred (Edge) | 0.0342 |
|                    | Diameter ( $D$ )                    | 6      |                   | High-pred (Node) | 0.0807 |
|                    | Sigma ( $\sigma$ )                  | 17.25  |                   | Low-pred (Edge)  | 0.4639 |
|                    |                                     |        |                   | Low-pred (Node)  | 0.9282 |

The resulting graph contains one single large connected component comprising of 2,617 nodes and 17,937 edges, corresponding to an average degree of 12.89 and a density of 0.0052. The low density indicates a sparse network, since only a small fraction of all possible user-user connections are realised. At the same time, the average degree shows that users are not isolated, but are embedded in a reasonably well-connected interaction structure. The network exhibits short global distances, with an average path length of 2.97 and diameter of 6, indicating that most users are separated by few interactions. Combined with a high small-world coefficient ( $\sigma = 17.25$ ), this suggests short paths alongside strong clustering relative to a random network. This is characteristic of small-world social networks, enabling efficient information flow within tightly connected local communities [31].

## Centrality Measures

The centrality statistics of this network, which were defined in Section 2.2.2, are displayed in Figure 5.1.

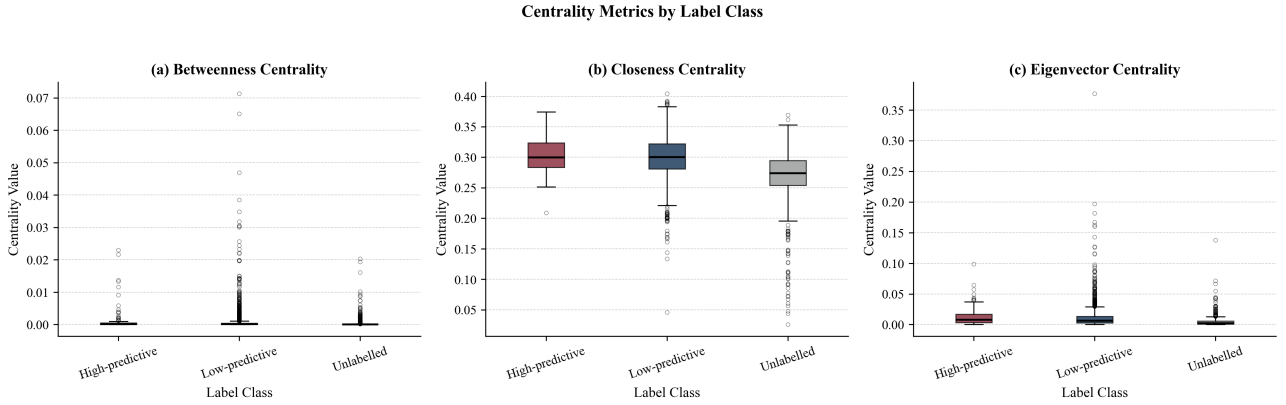


Figure 5.1: Betweenness (left), closeness (centre), and eigenvector (right) centrality by label class. High-predictive users show marginally higher closeness than unlabelled users but do not occupy broker positions, suggesting predictive ability is not strongly associated with structural centrality.

The mean closeness centrality is 0.29 (Table 5.1), indicating that nodes lie at relatively short geodesic distances from the rest of the network and are well-positioned for efficient information access. In contrast, mean betweenness centrality is substantially lower at 0.0009, suggesting that shortest-path brokerage is concentrated within a limited subset of users.

Disaggregating by label class (Figure 5.1), high-predictive users exhibit slightly higher median closeness than unlabelled users, though low-predictive users display comparable values. This pattern likely reflects a labelling artefact as users require sufficient posting or commenting activity to receive a label, which incidentally places them at more reachable network positions. Betweenness centrality remains near zero across all groups, indicating that brokerage roles are rare and not associated with predictive performance. Eigenvector centrality is similarly low and undifferentiated across groups since high eigenvector scores require connection to other high-scoring nodes, this indicates that high-predictive users are not embedded within mutually reinforcing clusters of highly active participants.

Predictive ability is therefore not strongly associated with occupying globally central, broker, or core-embedded positions, a finding that motivates the use of richer behavioural and local structural features in Chapter 6.

## Degree Distribution

The degree distribution shows how connections propagate across the graph.

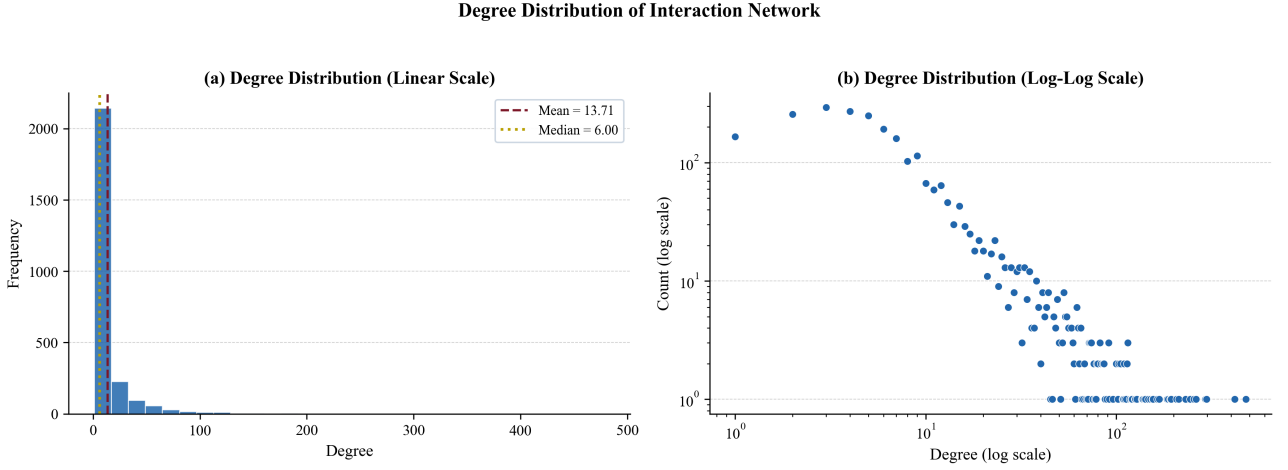
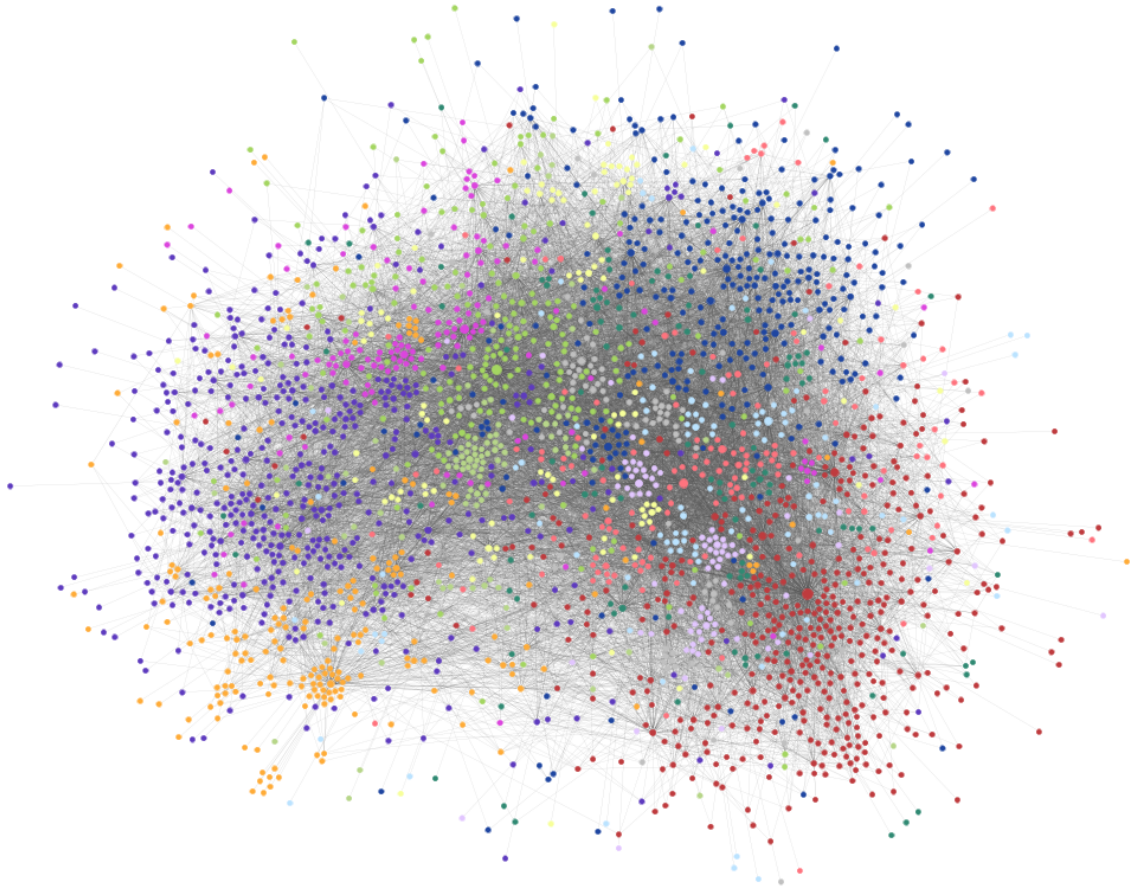


Figure 5.2: Degree distribution of the interaction network ( $n=2,617$ ) on linear (left) and log-log (right) scales. The right-skewed, heavy-tailed distribution is consistent with a scale-free network, where a small number of hub users reach degrees as high as 646.

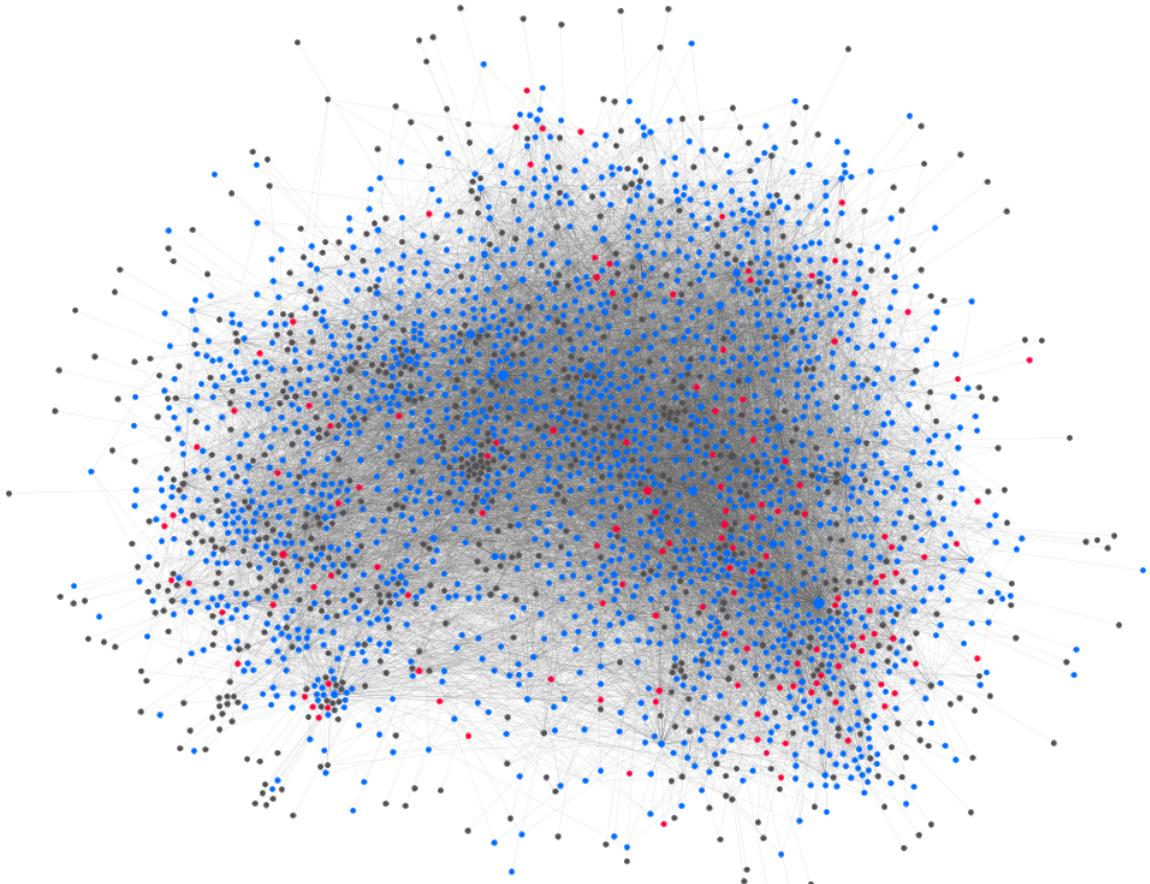
Figure 5.2 illustrates the degree distribution of the interaction network on both linear and log-log scales. The linear-scale histogram reveals a highly right-skewed distribution, where the vast majority of nodes have low degree, while only a small number of nodes exhibit high connectivity. The log-log plot reinforces this through an approximately linear decay in the tail of the distribution, consistent with a heavy-tailed degree distribution. Whether this follows a strict power law was not formally tested, but the qualitative pattern of a small number of high-degree hubs dominating an otherwise sparse network is characteristic of scale-free social networks [32]. While most user interactions are limited, a small number of hub nodes reach degrees as high as 646. However, since edge weights encode cumulative interaction frequency, a high unweighted degree does not necessarily imply broad reach, it could instead reflect repeated exchanges with a small set of users.

### 5.1.2 Community Structure

To examine the global structure of the network, community detection was performed using the Louvain algorithm. The resulting partition, illustrated in Figure 5.3a, identifies 14 distinct communities. While the network is highly interconnected without localised clustering, the communities appear distributed throughout the graph. For comparison, Figure 5.3b highlights the distribution of high-predictive and low-predictive users. In both figures, thicknesses of edges are proportional to the edge weight. From the figures, no significant spatial correlation between the Louvain communities and user predictability can be observed, as high-predictive users are dispersed across the entire network architecture rather than concentrated within specific clusters.



(a) Nodes coloured by Louvain communities (14 communities)



(b) Nodes coloured by predictive labels, (red = high-predictive, blue = low-predictive, grey = unlabelled). High-predictive users are dispersed across the network with no concentration in specific communities.

Figure 5.3: Full *r/CryptoMarkets* user interaction network (2,617 nodes, 17,937 edges).

## Modularity and Homophily

The observed modularity score of  $Q = 0.34$  (Table 5.1) indicates meaningful but not absolute separation between groups, and lies within the range commonly associated with non-random community structure in social networks [5]. The network remains strongly integrated within the modular organisation, as 67% of edges are inter-community links, indicating that a substantial proportion of interactions span community boundaries. Thus, the detected communities should not be interpreted as isolated subgraphs. Rather, they represent statistically cohesive groupings, nodes that interact more frequently with one another than expected under a random null model, while remaining structurally integrated through the 67% of edges that cross community boundaries. The network therefore exhibits modular organisation without sharp localised clustering.

The homophily was implemented as defined in Equation 2.3, and the resulting metric indicates strong label assortativity, with an average node homophily of 0.87, implying that users are frequently connected to others sharing the same predictive classification. This pattern is further reflected at the class level (Table 5.1) where low-predictive users exhibit high node-level homophily (0.93), whereas high-predictive users display substantially lower homophily (0.08). This asymmetry is consistent, and possibly driven by the underlying class imbalance (that will be discussed in Section 5.2), as the relatively small number of high-predictive users limits the formation of same-label neighbourhoods.

The degree distribution, modularity and small-world structure builds a case for the *r/CryptoMarkets* interaction graph being a recognisable modular, and yet still largely interconnected social structure. These properties are consistent with established characteristics of real-world social networks [31, 5], and provide a structural basis for investigation into the relationship between predictive capability and engagement across groups.

## 5.2 Analysis of Users

The labelling process in Section 4.6 yields 133 high-predictive users out of 2,617, a 13:1 class imbalance that defines the node labels for the graph machine-learning task. The skew is consistent with prior observations that informative forecasting in online financial communities is concentrated within a small subset of participants [3].

### 5.2.1 Analysis of User Sentiment

User sentiment, grouped by labels, is analysed in Figure 5.4, which shows that most are bullish, both among high-predictive and low-predictive groups, consistent with Figure 4.5 where bullish predictions are more numerous overall. Although bearish predictions appear to be more accurate on average, this does not translate into a predominantly bearish high-predictive group. This suggests that highly predictive users are not defined by a naive directional bias, but rather by an ability to identify either

market movement effectively.

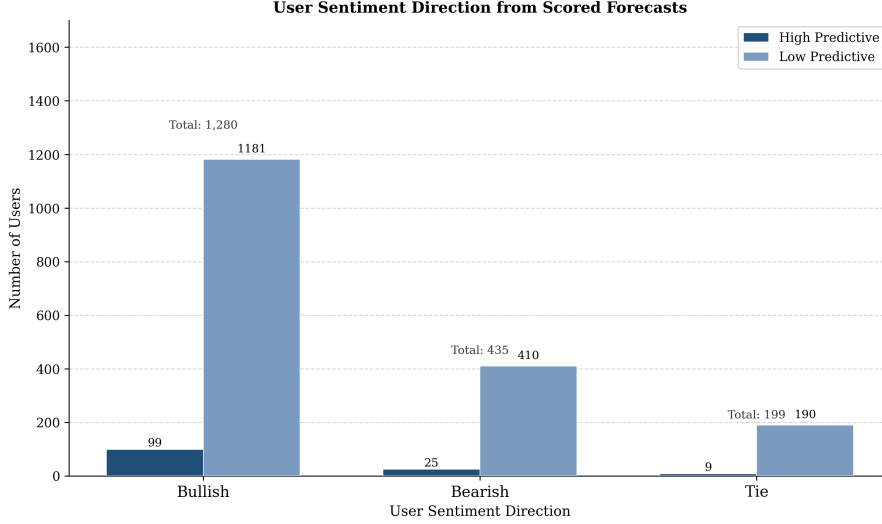


Figure 5.4: Dominant sentiment direction for high-predictive ( $n=133$ ) and low-predictive ( $n=1,781$ ) users. Both groups are predominantly bullish.

### 5.2.2 Analysis of User Attributes

The user attributes are represented as features. In total, there are 41 node features consisting of activity, forecast and structural attributes. Structural features are excluded from the MLP baseline to enable a controlled comparison between feature-only and graph-enhanced learning. A summary of all features is provided in Table A.1 and precise mathematical definitions in Appendix A.1.

Figure 5.5 compares the attributes of high-predictive users and low-predictive users, which might provide insight into possible beneficial node features in identifying users’ forecast abilities.

Despite the severe class imbalance, high-predictive users exhibit consistently stronger behavioural and structural signals. They show marginally higher mean degree ( $16.76$  vs.  $15.74$ ) and PageRank ( $4.27 \times 10^{-4}$  vs.  $4.45 \times 10^{-4}$ ). Unlike the centrality measures, that are global path-based, which showed no separation (Section 5.1.1), these local connectivity measures suggest a slight structural distinction, though the differences are modest. Behaviourally, they show a higher comment-to-post ratio, suggesting greater engagement in discussions rather than passive posting. They also exhibit marginally higher perceived confidence, but the similarity across classes indicates that confidence alone is not a reliable indicator of predictive ability. Finally, high-predictive users display a lower average post direction entropy, implying more consistent directional sentiment.

## 5.3 Labelled Social Interaction Graph

Figure 5.6 illustrates two local structural configurations. These are representative examples of the structural diversity in the graph rather than systematic characterisations of high-predictive users, whose identification requires integrating the full feature set described in Chapter 6.

### Predictive Label Analysis

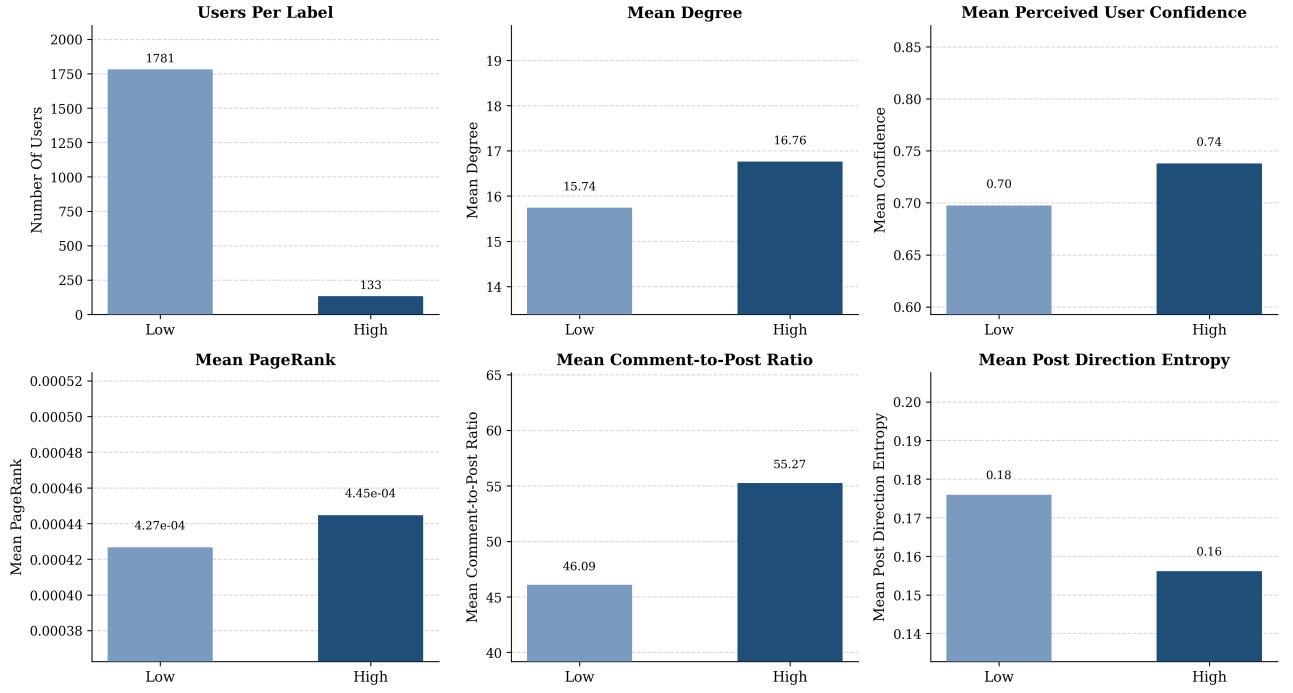
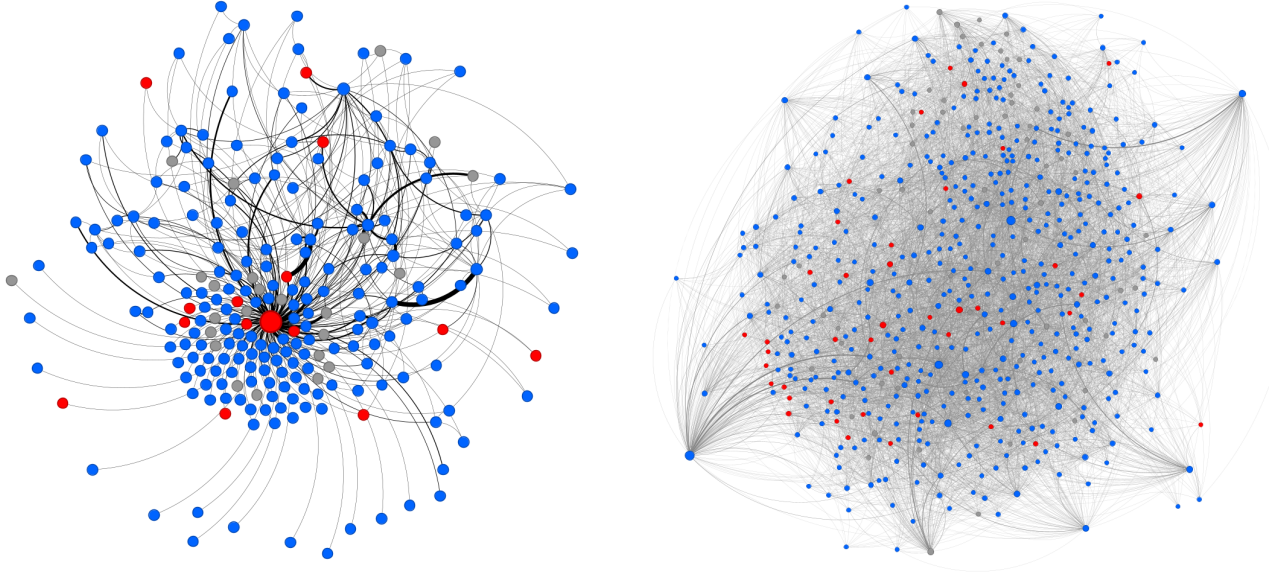


Figure 5.5: Mean node-level attributes for low-predictive (n=1,781) and high-predictive (n=133) users. Class imbalance is approximately 13:1.



(a) Ego network of radius 1 around a high-predictive hub, illustrating a potential structural hole brokerage role.

(b) 12-core subgraph showing the densely interconnected network backbone, where each node maintains at least 12 intra-core links.

Figure 5.6: Two subgraph views of the labelled network (red = high-predictive, blue = low-predictive, grey = unlabelled).

## 6 Graph Machine Learning

This chapter evaluates whether graph-based models can identify high-predictive users from the labelled social interaction graph  $\mathcal{G}$  constructed in Chapters 4. Labels  $\{\ell_u\}_{u \in \mathcal{V}_L}$  are available for only a subset of users, and the objective is to learn a classifier that generalises to unlabelled nodes  $\mathcal{V}_U$  under severe class imbalance (13:1) and heterophily ( $h = 0.08$  for high-predictive users). Additional definitions include:

1.  $H^{(l)} \in \mathbb{R}^{|\mathcal{V}| \times d_l}$  denotes the node representation at layer  $l$  with  $H^{(0)}$  as node feature matrix  $X$ .
2.  $h_u^{(l)} \in \mathbb{R}^{d_l}$  is the row of  $H^{(l)}$  corresponding to node  $u$ , so that  $H^{(l)} = [h_1^{(l)}, \dots, h_{|\mathcal{V}|}^{(l)}]^\top$ .
3.  $W^{(l)} \in \mathbb{R}^{d_l \times d_{l+1}}$  is the layer- $l$  learnable weight matrix.
4.  $\phi$  is defined as  $\phi(\cdot) = \text{Dropout}(\text{ReLU}(\text{Norm}(\cdot)))$ , except GAT which uses ELU in place of ReLU.

Eight architectures are compared to isolate the contribution of graph structure over feature-only and structure-only baselines: MLP, Label Propagation, GCN, GAT, GraphSAGE, H<sub>2</sub>GCN, MixHop, and a random baseline. The remainder of this chapter describes each architecture, the learning setup, and the evaluation objectives.

### 6.1 Comparative Model Architectures

While all architectures were grounded in its original formulation in the literature, the implementations follow practical adaptations provided by modern libraries (PyTorch Geometric), incorporating standard training components such as normalisation, dropout, and residual connections where appropriate [33]. Specifically, normalisation stabilises feature distributions across layers [34], dropout acts as a regulariser by randomly deactivating neurons during training [35, 36], and residual connections preserve node-specific information across layers, mitigating the loss of distinctive features caused by repeated neighbourhood aggregation [37]. Motivated by the low homophily of high-predictive users (Section 5.1.1), both homophily-based and heterophily-aware models are evaluated.

#### 6.1.1 Multi-layer Perceptron (MLP)

The Multi-layer Perceptron (MLP) isolates the predictive signal contained purely within node features, providing a baseline to assess whether graph structure contributes additional information. By ignoring relational dependencies encoded in the adjacency matrix, it assumes nodes are independent and identically distributed (i.i.d.) [36]. The MLP computes:

$$H^{(l+1)} = \phi(H^{(l)}W^{(l)}) \tag{6.1}$$

### 6.1.2 Label Propagation (LP)

Label Propagation (LP) provides a structure-only baseline by operating exclusively on the graph topology. Labels are iteratively propagated across edges based on neighbouring nodes, implicitly assuming homophily [38].

$$Y^{(t+1)} = \alpha \tilde{A} Y^{(t)} + (1 - \alpha) Y^{(0)}, \quad (6.2)$$

where  $Y^{(t)} \in \mathbb{R}^{|V| \times C}$  denotes the soft label matrix at iteration  $t$ , and  $C$  is the number of classes (here  $C=2$ ).  $Y^{(0)}$  is the initial label matrix with one-hot rows for labelled nodes and zero rows elsewhere.  $\tilde{A} = D^{-1/2} A D^{-1/2}$  is the symmetrically normalised adjacency matrix and  $\alpha \in (0, 1)$  controls the propagation strength.

### 6.1.3 Graph Convolutional Network (GCN)

The Graph Convolutional Network (GCN) extends neural networks to graph-structured data by aggregating features from local neighbourhoods via a normalised adjacency operator. In its original formulation, node representations are updated by combining transformed node features with those of their neighbours:

$$H^{(l+1)} = \text{ReLU} \left( \hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} H^{(l)} W^{(l)} \right), \quad (6.3)$$

where  $\hat{A} = A + I$  includes self-loops, ensuring that each node incorporates its own features during aggregation, and the normalised adjacency matrix  $\hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2}$  controls the contribution of neighbouring nodes [13]. This formulation can be interpreted as a neighbourhood averaging operation, where each node updates its representation based on a weighted combination of its own features and those of its immediate neighbours.

In this implementation, batch normalisation, dropout, and residual connections are incorporated:

$$H^{(l+1)} = \phi \left( \hat{D}^{-1/2} \hat{A} \hat{D}^{-1/2} H^{(l)} W^{(l)} \right) + H^{(l)} W_r^{(l)} \quad (6.4)$$

where  $W_r^{(l)} \in \mathbb{R}^{d_l \times d_{l+1}}$  is a learnable residual projection that aligns the input dimension with the output dimension.

### 6.1.4 Graph Attention Network (GAT)

Graph Attention Networks (GAT) extend GCN by introducing attention mechanisms to weight neighbour contributions [39], prioritising informative neighbours, which is beneficial in social networks where interactions vary in importance. The attention coefficient  $\alpha_{uv}$  between nodes  $u$  and  $v$  is computed as:

$$\alpha_{uv} = \frac{\exp \left( \text{LeakyReLU} \left( a^\top [h_u^{(l)} W^{(l)} \| h_v^{(l)} W^{(l)}] \right) \right)}{\sum_{k \in \mathcal{N}(u)} \exp \left( \text{LeakyReLU} \left( a^\top [h_u^{(l)} W^{(l)} \| h_k^{(l)} W^{(l)}] \right) \right)}, \quad (6.5)$$

where  $a \in \mathbb{R}^{2d_{l+1}}$  is a learnable attention vector,  $\parallel$  denotes concatenation, and the softmax normalisation is taken over all neighbours  $\mathcal{N}(u)$ . The attention-weighted adjacency matrix  $\tilde{A}_\alpha$  then has entries  $[\tilde{A}_\alpha]_{uv} = \alpha_{uv}$ .

Node representations are computed as:

$$H^{(l+1)} = \phi \left( \tilde{A}_\alpha H^{(l)} W^{(l)} \right) \quad (6.6)$$

In contrast to the original formulation which concatenates multi-head outputs, this implementation averages attention heads to maintain a fixed embedding dimension. Additionally, the activation function used is ELU rather than ReLU, as ELU produces smooth negative outputs rather than hard-zeroing them, which can improve gradient flow in attention-weighted aggregation. Accordingly, for GAT only,  $\phi(\cdot) = \text{Dropout}(\text{ELU}(\text{Norm}(\cdot)))$ .

### 6.1.5 Graph Sample and aggreGatE (GraphSAGE)

GraphSAGE learns node embeddings through sampled neighbourhood aggregation [11]. Rather than aggregating all neighbours, at each layer a fixed-size subset  $\tilde{\mathcal{N}}(u) \subseteq \mathcal{N}(u)$  is sampled uniformly, enabling inductive generalisation to unseen nodes. In this work, the mean aggregator variant is used:

$$h_u^{(l+1)} = \phi \left( \left[ h_u^{(l)} \parallel \frac{1}{|\tilde{\mathcal{N}}(u)|} \sum_{u \in \tilde{\mathcal{N}}(u)} h_u^{(l)} \right] W^{(l)} \right) + h_u^{(l)} W_r^{(l)}, \quad (6.7)$$

where  $\tilde{\mathcal{N}}(u)$  is the sampled neighbour set,  $\parallel$  denotes concatenation of the ego representation with the aggregated neighbourhood,  $W^{(l)} \in \mathbb{R}^{2d_l \times d_{l+1}}$  projects the concatenated vector to the next layer dimension, and  $W_r^{(l)} \in \mathbb{R}^{d_l \times d_{l+1}}$  is the residual projection.

### 6.1.6 H<sub>2</sub>GCN

H<sub>2</sub>GCN is designed for heterophilic graphs by explicitly separating ego-node features from neighbourhood representations and incorporating higher-order neighbourhood information [40]. In this work, a simplified H<sub>2</sub>GCN-inspired architecture is implemented:

$$H = \phi(XW_0) \parallel \phi(\hat{A}XW_1) \parallel \phi(\hat{A}^2XW_2), \quad (6.8)$$

where  $X \in \mathbb{R}^{|V| \times F}$  denotes the node feature matrix,  $\hat{A} = \hat{D}^{-1/2}(A + I)\hat{D}^{-1/2}$  is the symmetrically normalised adjacency with self-loops (as in GCN), and  $W_0, W_1, W_2 \in \mathbb{R}^{F \times d_{\text{out}}}$  are learnable projections for the ego, 1-hop, and 2-hop representations respectively. The terms  $\hat{A}X$  and  $\hat{A}^2X$  capture 1-hop and 2-hop neighbourhood features. By separating ego and neighbourhood representations, this formulation prevents the mixing of potentially dissimilar node features, while higher-order aggregation enables the

model to capture informative relationships beyond immediate neighbours.

### 6.1.7 MixHop

MixHop generalises graph convolutions by aggregating information from multiple powers of the adjacency matrix within a single layer [41]. By incorporating information from multiple hop distances, the model captures both local and higher-order structural patterns, enabling it to learn which neighbourhood scales are most informative. Node representations are updated as:

$$H^{(l+1)} = \left\|_{p \in \mathcal{P}} \left( \phi \left( \hat{A}^p H^{(l)} W_p^{(l)} \right) \right), \quad (6.9)$$

where  $\hat{A}^p$  denotes the  $p$ -th matrix power of the normalised adjacency  $\hat{A} = \hat{D}^{-1/2}(A + I)\hat{D}^{-1/2}$ ,  $\mathcal{P} = \{0, 1, 2\}$  is the set of propagation powers,  $W_p^{(l)} \in \mathbb{R}^{d_l \times d_{l+1}}$  is the per-power learnable weight matrix, and  $\|$  represents concatenation across different neighbourhood scales. The formulation allows the model to combine information from self-features ( $p = 0$ ), immediate neighbours ( $p = 1$ ), and more distant nodes ( $p = 2$ ), which is particularly beneficial in heterophilic networks where relevant signals may not be confined to local neighbourhoods. Unlike H<sub>2</sub>GCN, which concatenates ego and neighbourhood representations once using fixed 1-hop and 2-hop aggregations (Equation 6.8), MixHop applies a separate learnable weight matrix  $W_p^{(l)}$  per propagation power at every layer, allowing the model to learn which neighbourhood scales are most discriminative rather than treating all hops equally.

### 6.1.8 Random Baseline

A random baseline is included to establish a lower bound for performance. It provides a reference point to ensure that all learned models demonstrate predictive capability beyond chance. Predictions are generated independently for each node,  $\hat{y}_i \sim \text{Bernoulli}(0.5)$ , which corresponds to random assignment between the two classes with equal probability.

## 6.2 Learning Setup

The predictive models were implemented using PyTorch Geometric (PyG), chosen for its optimised handling of sparse graph structures [42], and executed across three random seeds (42, 100, 2026) to account for stochastic variation.

The labelled nodes were partitioned into train/validation/test sets using a stratified 60/20/20 split, ensuring proportional representation of the minority *High* class across all sets. Unlabelled nodes are excluded from the split but remain in the graph during training, providing structural context for neighbourhood aggregation. Node features were Z-score normalised prior to ingestion. Features derived from the labelling pipeline (`composite_score`, `accuracy`, `confidence_weighted`, `enhanced_weighted`) and metadata identifiers (`node_id`, `split`) were excluded to prevent data leakage, leaving 41 features as

described in Appendix A.1.

All neural models were optimised using Adam [43] with a class-weighted cross-entropy training loss to mitigate class imbalance. Checkpointing and early stopping were based on validation focal loss, with a patience of 50 epochs, a 10-epoch warm-up, and a minimum improvement threshold of  $1 \times 10^{-4}$ . Hyperparameters, namely hidden dimension, number of layers, dropout rate, and learning rate, were found through sensitivity tests across established ranges for GNN training [33]. After training, the operating threshold  $\tau$  is selected by a post-training sweep over candidate thresholds on the validation set. For each candidate, the recall of the positive class is computed and any threshold yielding recall below a minimum floor of 0.05 is rejected. Among the surviving candidates, the threshold with the highest positive-class precision is selected, giving the criterion: maximise **positive-class precision** subject to recall  $\geq 0.05$ . This mechanism is entirely separate from the loss functions used during training and validation as it operates post-training on fixed model outputs and involves no gradient computation.

Training, validation, and evaluation use different objectives to reflect the goals of each stage:

- **Training:** class-weighted cross-entropy loss
- **Validation:** focal loss (checkpointing and early stopping)
- **Evaluation:** AUROC and average precision (AP)

Full details are given in Sections 6.2.1 to 6.2.3.

### 6.2.1 Training Loss

The prediction task is a binary, class-imbalanced node-classification problem in which high-predictive users form the minority class. To reduce the bias of standard empirical risk minimisation towards the majority class, the training objective uses a **class-weighted cross-entropy loss**. For a labelled node  $i$  with true class  $y_i \in \{0, 1\}$  and predicted class probability  $p_{i,c}$  for class  $c$ , the loss is

$$\mathcal{L}_{\text{WCE}} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} \log p_{i,y_i}, \quad (6.10)$$

where  $w_{y_i}$  is the inverse-frequency weight associated with class  $y_i$  [36]. This increases the contribution of minority-class errors during optimisation and improves training stability under class imbalance. Focal loss is not used at the training stage, as it was seen to destabilise gradient updates in early epochs, possibly because unreliable probability estimates in early training caused the confidence-dependent modulation to suppress the wrong examples. Class-weighted cross-entropy provides a more consistent signal throughout optimisation.

### 6.2.2 Validation Loss

**Focal loss** was instead used for the validation objective, particularly checkpointing and early stopping, and it downweights well-classified examples and concentrates learning on harder cases. Let

$$p_t = \begin{cases} p, & y = 1, \\ 1 - p, & y = 0, \end{cases} \quad (6.11)$$

where  $p$  is the predicted probability of the positive class. The focal loss is

$$\mathcal{L}_{\text{FL}} = -\alpha(1 - p_t)^\gamma \log(p_t), \quad (6.12)$$

with focusing parameter  $\gamma = 2.0$  and scaling coefficient  $\alpha = 0.75$ . Focal loss applies a dynamic, confidence-based modulation, where examples with high  $p_t$  are down-weighted regardless of class, so a checkpoint with low  $\mathcal{L}_{\text{FL}}$  has specifically learned to handle uncertain cases near the decision boundary, a stronger proxy for generalisation than the training criterion alone [44].

Model outputs are continuous scores  $s(x)$  or positive-class probabilities, which are converted to hard predictions only when a decision threshold is required:

$$\hat{y}_\tau(x) = \begin{cases} 1, & s(x) \geq \tau, \\ 0, & s(x) < \tau. \end{cases} \quad (6.13)$$

The operating threshold  $\tau$  is tuned on the validation set by maximising **positive-class precision** subject to a minimum recall constraint, reflecting the need to prioritise reliable positive identifications while maintaining sufficient coverage.

### 6.2.3 Evaluation Metrics

The principal reported threshold-independent metrics are **average precision (AP)** and **area under the receiver operating characteristic curve (AUROC)**. Threshold-independent metrics are used for final evaluation because  $\tau$  is selected on the validation set. Reporting threshold-dependent metrics on the test set would conflate the choice of operating point with the quality of the learned score function. Average precision summarises precision-recall performance:

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n, \quad (6.14)$$

where  $P_n$  and  $R_n$  denote precision and recall at the  $n$ th threshold. This is the non-interpolated definition implemented in scikit-learn and is well suited to imbalanced settings, since precision-recall

analysis focuses directly on performance for the rare positive class [45, 46]. ROC analysis characterises classifier performance across all decision thresholds through the true positive rate

$$\text{TPR}(\tau) = \frac{\text{TP}(\tau)}{\text{TP}(\tau) + \text{FN}(\tau)}, \quad (6.15)$$

and false positive rate

$$\text{FPR}(\tau) = \frac{\text{FP}(\tau)}{\text{FP}(\tau) + \text{TN}(\tau)}. \quad (6.16)$$

The ROC curve is the set of points  $(\text{FPR}(\tau), \text{TPR}(\tau))$  traced as  $\tau$  varies, and the corresponding area under this curve is

$$\text{AUROC} = \int_0^1 \text{TPR}(u) du. \quad (6.17)$$

AUROC provides a threshold-independent measure of discrimination and may be interpreted as the probability that a randomly chosen positive instance is assigned a higher score than a randomly chosen negative instance [47]. AUROC and AP are reported together rather than individually because they capture complementary aspects of performance under class imbalance. AP penalises false positives heavily by focusing on the precision-recall trade-off, which is most relevant when acting on positive predictions is costly, whereas AUROC summarises global ranking quality across both classes. Due to AUROC being the standard evaluation metric in classification tasks [45, 47], AUROC is the primary evaluation metric.

In summary, this thesis uses class-weighted cross-entropy for optimisation, validation focal loss for checkpoint selection, and threshold-tuned positive-class precision to report the precision actually achieved at the specific operating threshold selected by the post-training sweep. AUROC and AP are the principal comparison metrics and are threshold-independent.

# 7 Experimental Results

## 7.1 Prediction Performance Metrics

### 7.1.1 Framework Results

Table 7.1 summarises the classification performance across all models. The primary evaluation metric is **AUROC**, as explained in Section 6.2. Ensembling across multiple random seeds stabilises predictions by averaging models trained from different initialisations, effectively mitigating variance arising from convergence to different local minima and improving generalisation. Hence, it usually has better performance than individual runs. Single-run in Table 7.1 refers to the model initiated with seed 42.

Table 7.1: Model performance summary. Bold indicates best performance per criterion.

| Framework          | Single-Run Prec. | Ens. Prec.   | Avg. Prec. (mean $\pm$ std)         | AUROC (mean $\pm$ std)              |
|--------------------|------------------|--------------|-------------------------------------|-------------------------------------|
| Random             | —                | —            | $0.087 \pm 0.000$                   | $0.498 \pm 0.000$                   |
| MLP                | 0.099            | 0.154        | $0.105 \pm 0.020$                   | $0.594 \pm 0.042$                   |
| LabelProp          | 0.094            | 0.123        | $0.096 \pm 0.001$                   | $0.583 \pm 0.005$                   |
| GCN                | 0.145            | 0.188        | $0.124 \pm 0.010$                   | $0.624 \pm 0.022$                   |
| GraphSAGE          | 0.120            | 0.235        | <b><math>0.140 \pm 0.002</math></b> | <b><math>0.632 \pm 0.016</math></b> |
| GAT                | <b>0.200</b>     | 0.112        | $0.098 \pm 0.019$                   | $0.575 \pm 0.068$                   |
| H <sub>2</sub> GCN | 0.000            | <b>0.300</b> | $0.110 \pm 0.013$                   | $0.609 \pm 0.039$                   |
| MixHop             | 0.051            | 0.170        | $0.100 \pm 0.027$                   | $0.561 \pm 0.095$                   |

Table 7.1 shows that GraphSAGE achieves the strongest overall performance, attaining the highest mean average precision ( $0.140 \pm 0.002$ ) and AUROC ( $0.632 \pm 0.016$ ). Relative to the random baseline, this represents improvements of 60.9% in AP and 26.9% in AUROC, and gains of 33.3% and 6.4% respectively over the feature-only MLP, confirming that combining node attributes with neighbourhood aggregation improves over both chance-level prediction and a feature-only baseline. Among the remaining models, GCN performs closest to GraphSAGE (AP:  $0.124 \pm 0.010$ , AUROC:  $0.624 \pm 0.022$ ). H<sub>2</sub>GCN achieves the highest ensemble precision (0.300), indicating it can achieve high precision. However it also seems to be less reliable as it suffered model collapse under seed 42. GAT achieves the highest single-run precision (0.200) but lower ensemble precision (0.112), suggesting instability across seeds. The structure-only Label Propagation and MixHop both remain close to random, with MixHop exhibiting the largest variance across seeds. Results that depend on operating thresholds are incredibly sensitive to individual predictions due to the small positive test set, and therefore should be interpreted with caution. Overall, GraphSAGE consistently offers the best balance of ranking performance and stability, and further analysis focuses on this model.

Figure 7.1 (left) shows the training and validation loss curves for the GraphSAGE model. The training loss decreases steadily throughout optimisation, indicating stable convergence and capability in the model’s ability in memorising the training data. The validation loss initially decreases rapidly,

### Training Results For GraphSAGE (Seed 42)

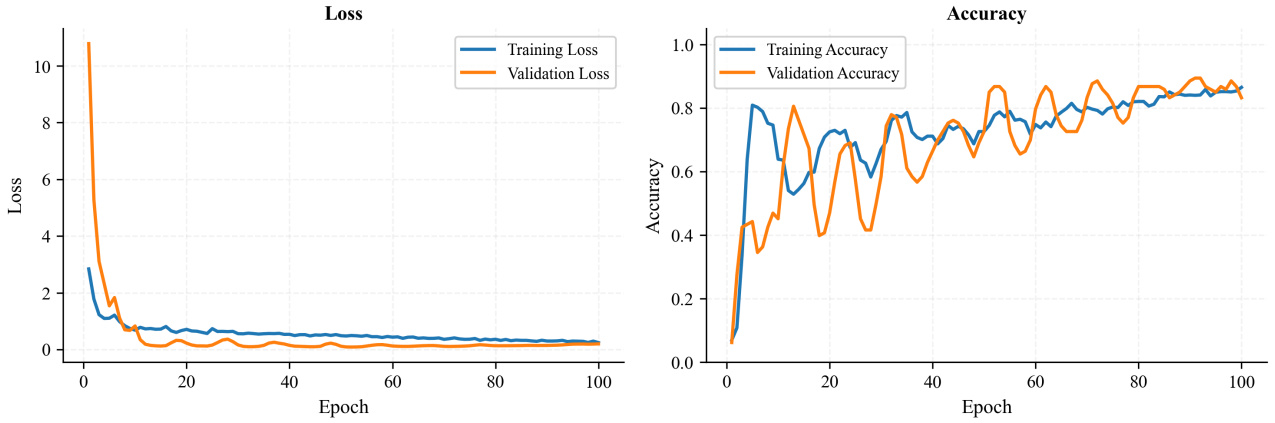


Figure 7.1: Training and validation loss (left) and accuracy (right) for GraphSAGE (seed 42). Validation loss stabilises before rising slightly, indicating mild overfitting mitigated by early stopping. Both accuracy curves converge to approximately 0.82.

suggesting effective early learning, before stabilising and exhibiting a slight upward trend. This indicates the onset of mild overfitting, where the model continues to improve on the training data while generalisation performance plateaus. The use of early stopping mitigates this by storing the model parameters at the point of lowest validation loss. The accuracy curves (Figure 7.1, right) show high initial volatility in both training and validation accuracy, with both curves converging to approximately 0.82. The initial volatility in validation accuracy reflects the small validation set and class imbalance, but the subsequent stabilisation indicates consistent generalisation performance.

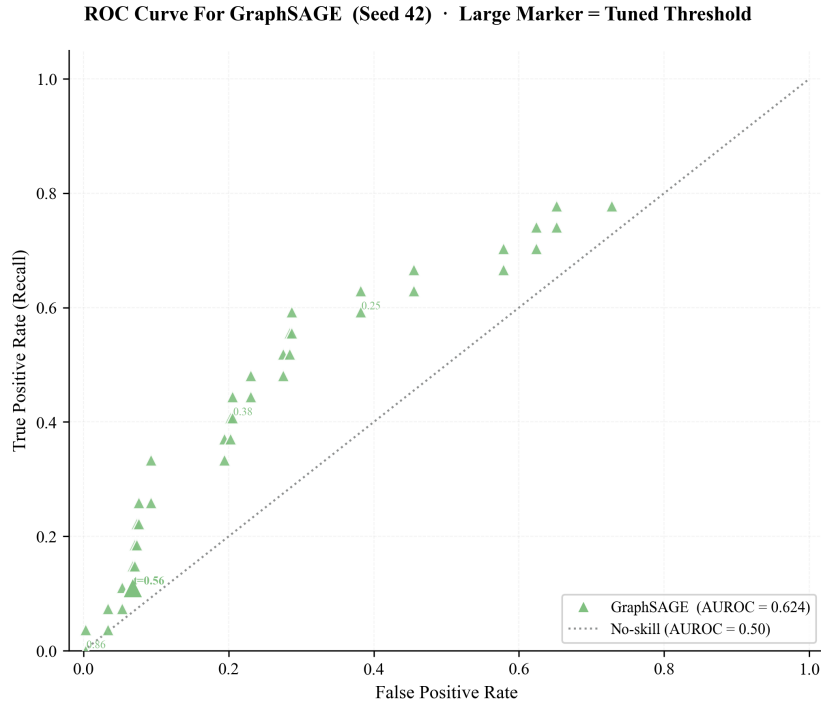


Figure 7.2: ROC curve for GraphSAGE (seed 42) (AUROC = 0.624) against the no-skill baseline (AUROC = 0.50). The gradual improvement as the threshold relaxes indicates moderate rather than strong class separation. The large marker denotes the operating threshold of 0.56.

Figure 7.2 shows the ROC curve for the GraphSAGE model under seed 42. The curve lies consistently above the no-skill baseline, reinforcing the AUROC results in Table 7.1 that the model is able to distinguish between high-predictive and low-predictive users better than random chance. However, the curve does not exhibit strong convexity towards the top left corner. This is in contrast to an ideal ROC profile of a near-perfect classifier, where a curve rises steeply towards a true positive rate of 1 before the false positive rate increases. Instead, performance improves gradually as the decision threshold is relaxed, indicating a trade-off between true positive rate and false positive rate across thresholds. The operating threshold of 0.56 reflects the precision-oriented threshold tuning strategy, which prioritises high-confidence predictions over maximising coverage.

### 7.1.2 Effect of Labelling Criteria on Model Performance

To assess the sensitivity of results to the user labelling threshold, two variants of the labelling pipeline were compared and recorded in Table 7.2. The standard criteria, adopted throughout this thesis, assigns the high-predictive label using a composite score cutoff of 0.66. A softened variant was also evaluated by relaxing this cutoff to 0.5, which increases the number of positive labels and thereby reduces the severity of class imbalance.

Table 7.2: Model performance under standard and softened labelling criteria. All metrics reported as mean  $\pm$  std across seeds 42, 100, and 2026. Bold indicates best performance per criterion.  $h$  denotes graph homophily.

| Framework              | Softened Criteria ( $h = 0.57$ )    |                                     | Standard Criteria ( $h = 0.87$ )    |                                     |
|------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-------------------------------------|
|                        | AUROC                               | Avg. Precision                      | AUROC                               | Avg. Precision                      |
| Random                 | 0.497                               | 0.384                               | 0.498                               | 0.087                               |
| MLP                    | $0.519 \pm 0.031$                   | $0.404 \pm 0.012$                   | $0.594 \pm 0.042$                   | $0.105 \pm 0.020$                   |
| LabelProp <sup>†</sup> | <b><math>0.586 \pm 0.003</math></b> | $0.453 \pm 0.007$                   | $0.583 \pm 0.005$                   | $0.096 \pm 0.001$                   |
| GCN                    | $0.527 \pm 0.003$                   | $0.443 \pm 0.007$                   | $0.624 \pm 0.022$                   | $0.124 \pm 0.010$                   |
| GraphSAGE <sup>‡</sup> | $0.531 \pm 0.020$                   | $0.435 \pm 0.012$                   | <b><math>0.632 \pm 0.016</math></b> | <b><math>0.140 \pm 0.002</math></b> |
| GAT                    | $0.548 \pm 0.012$                   | $0.427 \pm 0.019$                   | $0.575 \pm 0.068$                   | $0.098 \pm 0.019$                   |
| H2GCN                  | $0.538 \pm 0.047$                   | $0.445 \pm 0.038$                   | $0.609 \pm 0.039$                   | $0.110 \pm 0.013$                   |
| MixHop                 | $0.566 \pm 0.024$                   | <b><math>0.483 \pm 0.028</math></b> | $0.561 \pm 0.095$                   | $0.100 \pm 0.027$                   |

<sup>†</sup> Best-performing model under softened criteria according to AUROC.

<sup>‡</sup> Best-performing model under standard criteria according to AUROC.

The key structural effect of relaxing the threshold is a reduction in graph homophily from 0.87 to 0.57, which highlights how the high homophily was largely driven by the majority low-predictive class. Under the standard criteria, the small number of high-predictive users are scattered across the network with node-level homophily of just 0.08, while the dominant low-predictive class achieves homophily of 0.93. Label Propagation can therefore propagate labels effectively for the majority class but fails on the minority class, where same-label neighbourhoods are too sparse to support smoothing. Relaxing the threshold increases the prevalence of the high-predictive class, giving its members enough same-label neighbours for the smoothness assumption underlying label propagation to hold for both classes simultaneously. This explains why Label Propagation now achieves the strongest AUROC (0.586)

under the softened criteria, underlining that graph topology contains informative signals, though it is a modest improvement from the random baseline.

The random baseline achieves an average precision of 0.384, compared to 0.087 under the standard criteria, reflecting a substantially higher prevalence of positive labels. This also inflates the apparent performance of weaker models such as MLP, which achieves an average precision of 0.404 under softened criteria compared to 0.105 under standard criteria, making cross-criteria comparisons unreliable without baseline normalisation.

Overall, these results indicate that there are structural signals that encapsulate the node’s predictive ability when the threshold is perturbed. The comparative advantage of graph-based models is most clearly demonstrated under the standard, harder labelling regime. In particular, the superiority of GraphSAGE over both feature-only and structure-only baselines is more pronounced when class imbalance is severe and the graph is heterophilic. The primary analysis continues with the standard criteria to remain faithful to the definition of the prediction task itself.

### 7.1.3 Misclassification Analysis

Error analysis was conducted to identify the sources of misclassification. The confusion matrix in Figure 7.3 provides further insight into GraphSAGE’s classification behaviour.

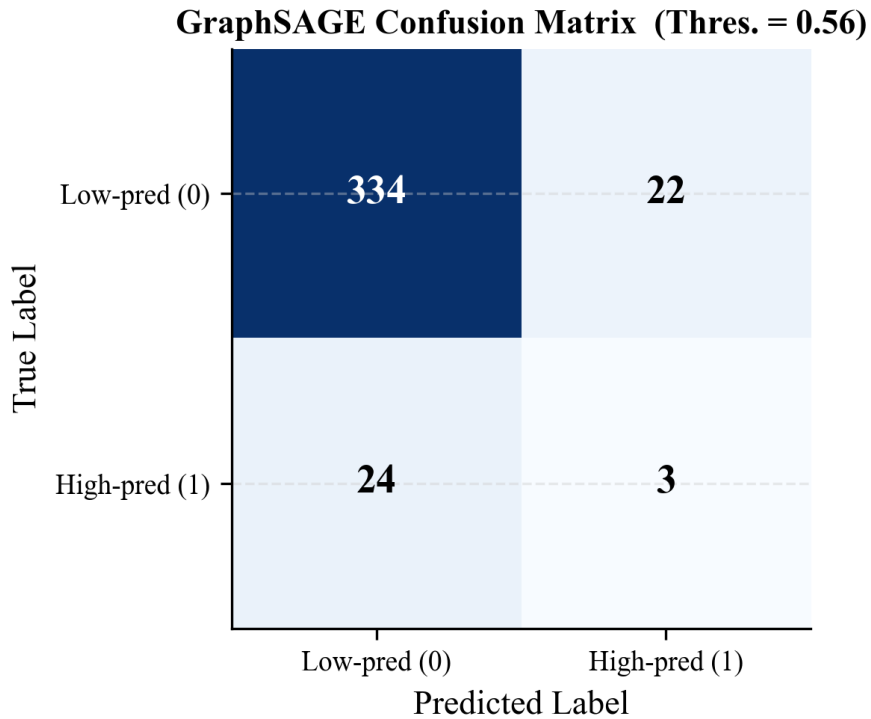


Figure 7.3: GraphSAGE (seed 42) confusion matrix. The model identifies 3 true positive high-predictive users against 22 false positives and 24 false negatives, yielding approximately 12% precision. Strong majority-class performance (334 true negatives) reflects the 13:1 class imbalance.

The model correctly identifies 3 high-predictive users (true positives) while incorrectly labelling 22 low-predictive users as high-predictive (false positives), resulting in a precision of approximately 12%.

In contrast, performance on the majority class is considerably stronger, with 334 true negatives and only 24 false negatives. This reflects the underlying class imbalance and the difficulty in accurate identification of the minority class. The small test set means each individual prediction carries significant weight, and therefore precision will be highly sensitive. Nevertheless, examining the characteristics of each group reveals coherent and interpretable error patterns. Table 7.3 summarises the prediction statistics for each outcome group. Table 7.4 summarises the structural metrics and scores across the three groups and Table 7.5 details the five most able missed users.

Table 7.3: Prediction statistics by classification outcome. Avg. Std. Acc. is the per-user mean of the directional-correctness indicator. Avg. Enh. Acc. uses the stricter criterion in Section 4.6.1. Avg. Conf. is the mean perceived confidence across the user’s directional forecasts.

| Outcome        | Users | Forecasts | Scored | Avg. Std. Acc. | Avg. Enh. Acc. | Avg. Conf. |
|----------------|-------|-----------|--------|----------------|----------------|------------|
| True positive  | 3     | 63        | 24     | 79.2%          | 66.7%          | 0.730      |
| False positive | 22    | 958       | 423    | 40.2%          | 25.6%          | 0.714      |
| False negative | 24    | 469       | 185    | 83.2%          | 63.6%          | 0.719      |

Table 7.4: Mean structural features and scores by classification outcome. Composite Score combines directional accuracy, confidence-weighted accuracy, and enhanced accuracy (Equation 4.10). Prob. Score is the GraphSAGE output score prior to thresholding.

| Outcome        | Degree | Wtd. deg. | PageRank | Composite Score | Prob. Score |
|----------------|--------|-----------|----------|-----------------|-------------|
| True positive  | 9.0    | 19.7      | 0.000318 | 0.740           | 0.686       |
| False positive | 29.9   | 50.8      | 0.000781 | 0.425           | 0.682       |
| False negative | 16.9   | 25.8      | 0.000424 | 0.734           | 0.277       |

Table 7.5: Top five false negative users ranked by composite score. These high-predictive users received model probability scores below the decision threshold of  $\tau = 0.56$ . User identifiers are anonymised in accordance with the ethical considerations (Section 8.3.2)

| User | Composite | Pred. acc. | Degree | Prob. score | Inferred reason for miss                                                           |
|------|-----------|------------|--------|-------------|------------------------------------------------------------------------------------|
| FN-1 | 0.923     | 92.9%      | 5      | 0.381       | Low structural prominence                                                          |
| FN-2 | 0.828     | 88.9%      | 111    | 0.059       | High degree but near-zero probability score; majority-class hub neighbour dilution |
| FN-3 | 0.820     | 75.0%      | 42     | 0.491       | Borderline probability score                                                       |
| FN-4 | 0.815     | 100%       | 12     | 0.134       | Low structural prominence                                                          |
| FN-5 | 0.808     | 100%       | 14     | 0.151       | Low structural prominence                                                          |

**True Positives** The three correctly identified users share a profile of moderate activity, low structural prominence, and high prediction scores. Their mean degree is 9.0 and mean pagerank is 0.000318, placing them in relatively peripheral network positions. Their directional forecast accuracy is 79.2%, and decreasing to 66.7% under the enhanced criterion.

**False Positives** The 22 false positive users are structurally prominent but predictively weak. Their mean degree is 29.9, which is more than three times that of the true positives. Their mean pagerank (0.000781) is more than double. Despite this, their directional forecast accuracy is only 40.2%, barely above the random baseline for this prevalence level, and their enhanced accuracy drops to 25.6%.

**False Negatives** The 24 missed users represent the most consequential error class. Their mean composite score is 0.734, nearly identical to that of the true positives (0.740), and their directional forecast

accuracy of 83.2% actually exceeds both other groups. Despite this, their mean model probability score is 0.277, far below the decision threshold of 0.56.

Table 7.5 illustrates the individual patterns through the five highest-composite FN users. Graph aggregation will be less effective when minority nodes have few neighbours (FN-1, FN-4, FN-5) or are surrounded by majority-class hubs (FN-2), both of which are compounded by the 13:1 class imbalance leaving too few same-label neighbours to support reliable message-passing. Further analysis can be done to confirm if this pattern generalises across seeds and data splits.

## 7.2 Robustness Checks

### 7.2.1 Ablation Tests

In order to understand the individual contributions of all 41 features, an ablation test was conducted on GraphSAGE. Each feature was removed to see how the classification metrics changed with and without each feature. The 5 most beneficial and harmful features were plotted in Figure 7.4.

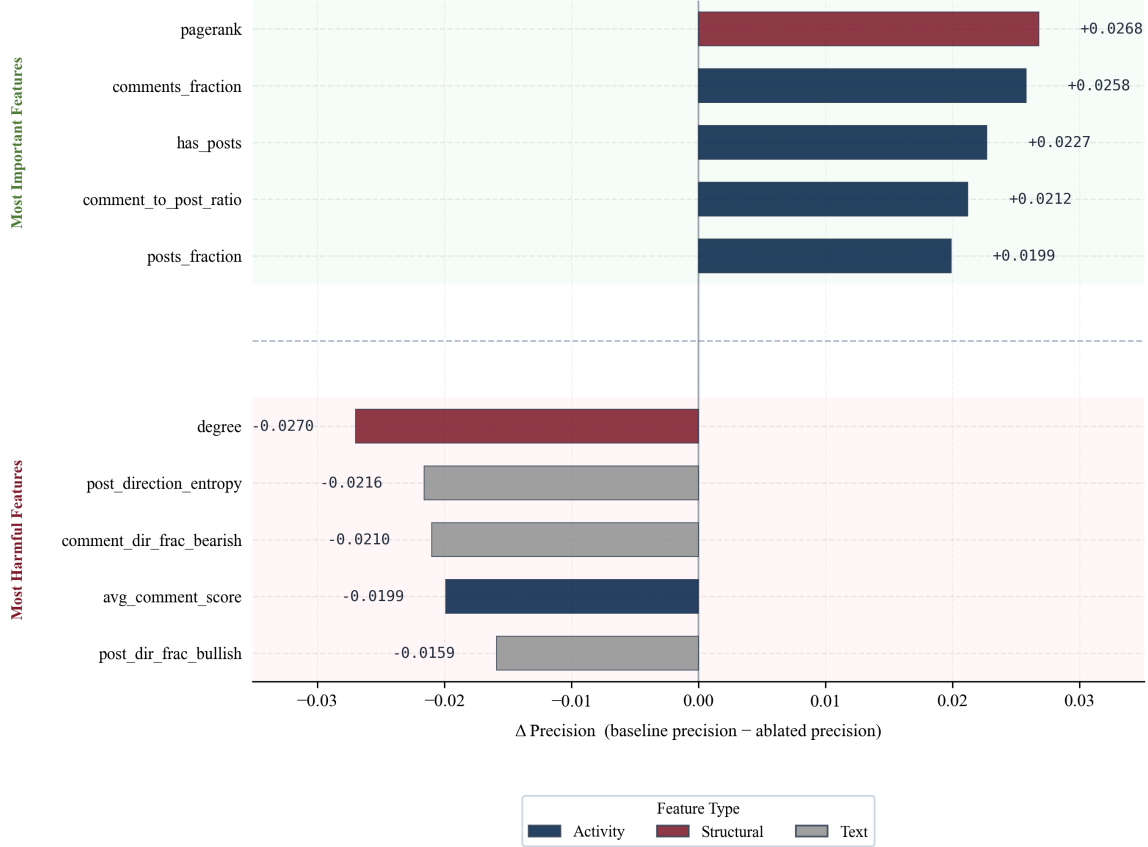


Figure 7.4: Ablation study on GraphSAGE (seed 42) showing the change in average precision for the top 5 most beneficial and harmful features.

Figure 7.4 highlights the relative importance of both structural and behavioural features in this classification setting. Among the most influential features, **pagerank** produces the largest positive impact, with average precision dropping 0.0268 when removed. This is closely followed by activity-based features such as **comments\_fraction** (0.0258), **has\_posts** (0.0227), and **comment\_to\_post\_ratio** (0.0212),

suggesting that user engagement patterns are also highly informative. Contrary to expectations, not all structural features are beneficial. While `pagerank` contributes positively, the removal of `degree` results in an increase in precision of 0.0270, indicating that it a harmful feature in this setting.

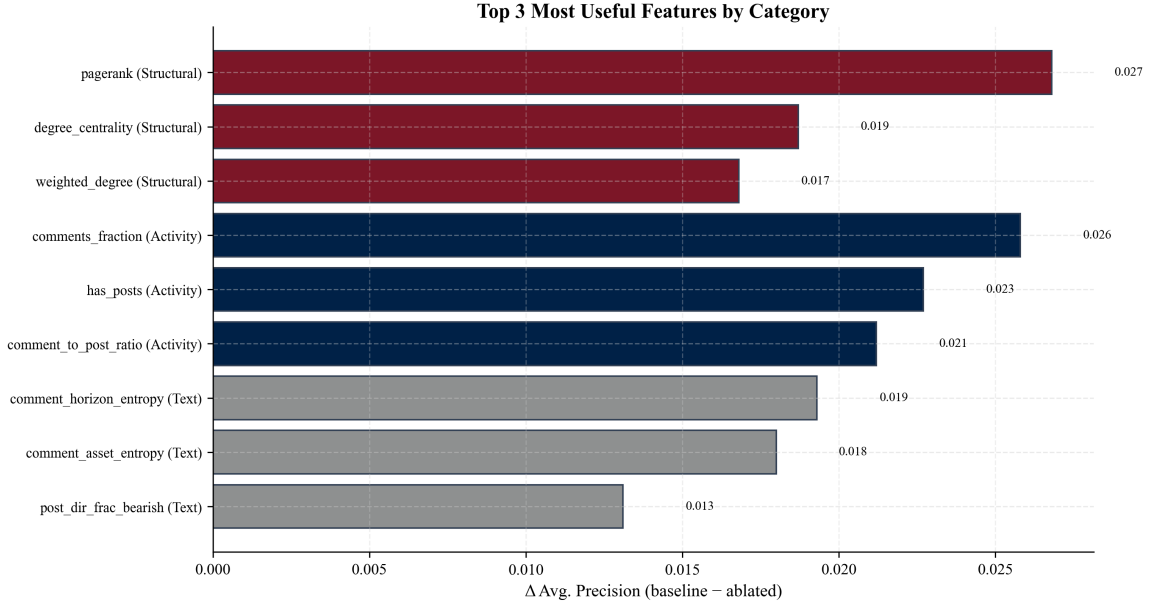


Figure 7.5: Top three most informative features per category from the GraphSAGE (seed 42) ablation study, measured by the drop in average precision when each feature is individually removed. Larger values indicate greater contribution to model performance.

To contextualise these findings further, the three most informative features within each category (structural, activity and text) are summarised from the ablation analysis and displayed in Figure 7.5. Among structural features, `pagerank` produces the largest average precision drop when removed (0.027), followed by `degree centrality` (0.019) and `weighted degree` (0.017). The consistent contribution of all three suggests that influence within the network, measured across multiple complementary definitions, is a partial predictor of high-predictive behaviour. The top three activity features (`comments fraction`, `has posts`, `comment to post ratio`) collectively capture the balance of a user’s participation across submission and commentary roles, rather than the absolute volume of activity. Text features seem to have the least contribution, with `comment horizon entropy` (0.019), `comment asset entropy` (0.018), and `post direction fraction bearish` (0.013) having the highest positive impact.

### Interpreting Ablation Results

Single-feature ablation provides a useful but partial view of feature importance. Because features are removed independently, the reported drops do not capture interactions between correlated features. Activity features such as `comments fraction`, `comment to post ratio`, and `has posts` encode overlapping aspects of participation style, so individual contributions are likely underestimated when the others remain present. This distinction between marginal and collective contribution is a known

limitation of permutation-style importance methods [48].

Threshold-dependent metrics also introduce sensitivity under class imbalance. Because the decision threshold is returned after each feature removal and the validation set contains few positive instances, precision estimates can shift meaningfully with single prediction changes [46]. Features with small reported drops should therefore be treated as indicative rather than precise, and cross-feature comparisons within the same order of magnitude should not be over-interpreted.

With these methodological caveats in mind, the broad patterns emerging from the ablation study converge on a coherent behavioural profile that high-predictive users generally occupy structurally influential positions, engage actively and responsively rather than broadcasting in volume (ablation results of `pagerank` and `comments_fraction` (Figure 7.4) aligns with the user attribute analysis (Section 5.2.2)). Textual features having the least association with model performance (Figure 7.5) might indicate that there is a large variety in direction of forecasts among high-predictive users despite such users having a lower average `post_direction_entropy` (Section 5.2.2). Considering that bearish forecasts make up only 9% of all content (Section 4.4.3) but achieve higher directional accuracy than bullish ones (Section 4.5.4), the results might suggest that there is a large variety of opinions of high-predictive users and actually shows the variety of high-predictive users, some might be more directionally consistent, and some might be able to identify and predict both bullish and bearish market movements.

### 7.2.2 Graph Topological Perturbation Tests

To quantitatively understand the impacts of the structural features in greater detail, random perturbations and DICE perturbations were conducted on GraphSAGE. The random perturbation tests randomly corrupts the graph structure and determines the degree of reliance on the graph structure. DICE takes this one step further, replacing intra-class edges with inter-class edges. The degree of each node was preserved in this study. The model was trained on the clean graph, and retained for the increasingly perturbed graphs. The model’s mean and standard deviation of the precision was recorded in Figure 7.6. Unlike the other evaluations, the perturbation tests record test accuracy rather than precision. It was observed that the precision fluctuated substantially with individual prediction changes, making it difficult to observe a clean trend. Accuracy across all nodes provides a smoother signal for visualising the monotonic effect of structural degradation, and is used here for illustrative purpose rather than as a performance claim.

Under random perturbations, model performance degrades steadily, confirming that the graph structure contains informative relational signals. Despite the extreme perturbation on the graph, the test accuracy only decreases marginally from approximately 0.87 to 0.83, which might indicate a limited

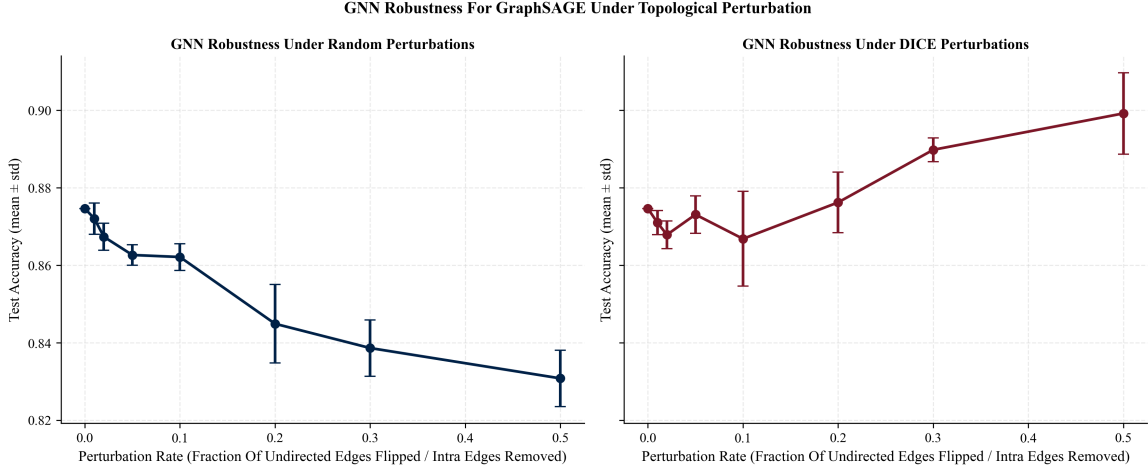


Figure 7.6: GraphSAGE (seed 42) test accuracy under random (left) and DICE (right) topological perturbations at increasing perturbation rates, with mean and standard deviation across seeds. Random perturbations degrade performance whereas DICE perturbations improve performance

structural signal, or is an observed effect of severe class imbalance.

However, performance unexpectedly improves from 0.87 to 0.89 under DICE perturbations. This could be due to the strong homophily of the graph (Table 5.1). The dense same-label connections might not provide discriminative contrast as majority-class nodes form tightly homogeneous neighbourhoods that dilute the minority signal during aggregation. DICE inadvertently corrects this structural bias by rebalancing edges toward the cross-class interaction pattern the model needs to exploit, potentially removing structural noise while retaining the informative signal.

Nevertheless, the fact that an edge-rewiring heuristic improves performance suggests that GraphSAGE does not fully exploit the heterophilic structure under its standard mean aggregation scheme, which implicitly assumes neighbouring nodes carry consistent signals, an assumption that holds poorly when high-predictive users are predominantly surrounded by low-predictive neighbours.

Heterophily-aware architectures such as H<sub>2</sub>GCN are in principle better suited to this setting, though their failure to consistently outperform GraphSAGE here likely reflects the compounding difficulty of severe class imbalance that were not fully addressed by either architecture. H<sub>2</sub>GCN separates ego and neighbourhood representations to avoid mixing dissimilar features, but when the minority class constitutes only 6.9% of nodes, the ego-neighbourhood contrast it relies on is too sparse to propagate a reliable signal. GraphSAGE succeeds not because it handles heterophily better in principle, but because its inductive neighbourhood sampling is more robust to the compounding difficulty of sparse positive labels and noisy cross-class edges and the strength of classic graph neural networks when hyperparameters are properly tuned [33]. This suggests that future architectures for this setting should address class imbalance and heterophily simultaneously rather than treating them as independent problems.

## 8 Discussion

### 8.1 Key Features Driving Predictions

The ablation results (Figure 7.4) show that GraphSAGE is driven by a mixture of behavioural, structural, and textual features, with behavioural signals contributing most strongly on average. In particular, features that are related to the type of subreddit activity such as `comments_fraction`, `has_posts`, `comment_to_post_ratio`, and `posts_fraction` are among the most beneficial features, suggesting that the manner in which users participate in discussions is more informative than raw activity volume alone. This is consistent with the user-level analysis (Figure 5.5), where high-predictive users exhibit engagement with other content.

These findings suggest that the model is capturing meaningful behavioural patterns rather than relying on superficial signals. However, they should not be interpreted as causal. Strong engagement and directional consistency may simply proxy user experience or community embeddedness rather than intrinsic forecasting ability. They may also partly reflect the Bayesian shrinkage procedure described in Section 4.6.2, which rewards sustained predictive consistency. The model may therefore be learning signals associated with user visibility and stability as well as predictive skill.

The role of structural features is more mixed. On the one hand, `pagerank` is the single most important feature, with its removal decreasing average precision by 0.0268, which shows that the model might be using this metric to identify true positives, to suppress false positives, or both. Since `pagerank` depends on the importance of neighbouring nodes as well as the number of connections, it captures a more meaningful notion of influence than simple activity count. On the other hand, removing `degree` increases precision by 0.0270, suggesting that coarse connectivity measures may introduce noisy information. This is plausible given that GraphSAGE already incorporates graph topology and explicit structural node features may duplicate information already encoded by GNN message passing, or introduce noisy connectivity signals that hurt discrimination. Additionally, the maximum change in average precision observed during the ablation study is 22.5%, which is substantial. This highlights both the extent to which predictive performance depends on the combined contribution of multiple features and the fact that certain individual features might carry particularly strong discriminative signal.

Among the architectures evaluated, GraphSAGE achieves the strongest balance of feature and structural signal under heterophily and severe class imbalance, outperforming both the feature-only MLP and the structure-only Label Propagation baseline. This pattern indicates that the predictive signal lies in the interaction between node attributes and local relational context rather than in either alone,

and that GraphSAGE’s mean aggregation over sampled neighbourhoods is robust enough to extract this signal without being destabilised by noisy or class-imbalanced edges. The present task is formulated as semi-supervised transductive learning, in which test nodes are present in the graph during training, so GraphSAGE’s inductive capability is not directly exploited. It is, however, a practically relevant property for future deployment, where the model could classify newly joined users by aggregating features from their emerging neighbourhoods without retraining, making it a natural candidate for real-time identification pipelines in dynamic social networks.

## 8.2 Implications for Cryptocurrency Communities

The user labelling pipeline developed in Chapter 4 is itself a methodological contribution independent of the graph learning results. Identifying high-predictive users in online financial communities requires resolving several non-trivial steps, including extracting directional forecasts from unstructured text, assessing forecast outcomes against price data, and aggregating individual prediction scores into a stable user-level label under class imbalance. The composite scoring approach provides a principled and replicable framework for this task. The use of GPT-4o-mini for forecast extraction further demonstrates that large language models can serve as scalable annotation tools in financial social media settings. This pipeline is not specific to *r/CryptoMarkets* and could be applied to other online communities where users make verifiable directional claims, such as stock discussion forums or prediction markets.

These findings reinforce that useful predictive information is not evenly distributed across the cryptocurrency discussion network. Instead, it is marginally associated with users who occupy influential positions and display distinctive engagement patterns (Section 5.2.2). This is consistent with the positive role of **pagerank** and with the superior performance of GraphSAGE over both the feature-only MLP and the structure-only Label Propagation baseline. Graph structure matters, but only when interpreted alongside node-level behavioural information. Furthermore, the combination of ablation and perturbation results indicates that crypto discussion networks contain both signal and noise. Certain behavioural and influence-based structural features help identify predictive users, yet some explicit structural features and some observed edges reduce performance. Information flow in these communities is therefore neither purely random nor adequately described by simple homophily. Rather, it reflects a heterogeneous mixture of informative interactions, noisy engagement, and redundant connectivity.

The practical implication is that identifying high-quality information sources in online financial communities requires models that can exploit local graph structure while remaining robust to noisy topology. GraphSAGE outperformed baselines because it combines informative node features with

neighbourhood context, but the perturbation results show that even this model benefits when edges are removed, possibly because a subset of observed edges carry noisy or misleading structural signal, though whether individual edges are harmful cannot be directly verified from perturbation experiments alone. Expert identification in crypto communities should therefore rely not on popularity alone, but on how users participate, where they sit in the network, and how information propagates through the broader interaction structure.

Together, the user-labelling pipeline, the architecture comparison, and the ablation and perturbation analyses provide a coherent framework for locating and characterising informative participants in online financial communities.

## 8.3 Limitations and Ethical Considerations

### 8.3.1 Limitations

Although *r/CryptoMarkets* provides a substantial quantity of user-generated content, signals are limited to a single subreddit. Users classified as high-predictive here may exhibit distinctive patterns across other cryptocurrency communities, and future work could examine whether identifiable structural characteristics persist across multiple subreddits, incorporating cross-community behaviour.

A fundamental limitation is noise in the labelling pipeline itself. GPT-4o-mini achieves a macro-F1 of 0.78 and Cohen’s kappa of 0.76 on forecast extraction, however approximately 15-20% of forecasts may be incorrectly labelled. These misclassified forecasts were often at the boundary between forecasts and non-forecasts, and therefore do not indicate a global misclassification problem. Nonetheless, this error can propagate through the user scoring and the training labels, potentially misclassifying users at the boundary of the high-predictive tier. Future work should quantify this sensitivity explicitly.

A further source of bias is post-hoc curation by users. Reddit allows authors to delete or edit their submissions, and users may selectively remove forecasts that proved incorrect, whether to curate their public image or to avoid embarrassment. Pushshift’s archival shortly after submission mitigates the impact of deletions occurring after the archival window, but content removed before archival would not appear in the dataset at all. To the extent that this curation is non-random, measured forecast accuracy among such users may be systematically inflated, biasing the high-predictive label toward those who curate their history rather than those who genuinely forecast well. A sensitivity check stratifying performance by the edit-status field already present in the dataset (Section 4.2.1) would partially address this concern. A more thorough analysis would require comparing the Pushshift snapshot against the live Reddit API to identify post-archival deletions.

The price data introduces a further source of label noise. Evaluation relied on endpoint price movements from *yfinance*, so intra-period volatility was not captured. Higher-resolution financial data

would allow a more precise assessment of forecast quality and more reliable user labels. The temporal scope is also a constraint. Extending the observation window across different market regimes would clarify whether predictive ability is stable over time or contingent on short-term conditions. Finally, annotation bias from a single annotator could be mitigated in future work through multiple annotators with clear inter-annotator agreement protocols.

### **8.3.2 Ethical Considerations**

An important ethical consideration in this study concerns the privacy of Reddit users, since user identifiers were collected and analysed as part of the network construction process. This concern is legitimate given the broader public sensitivity surrounding personal data use. However, the data used in this project were drawn exclusively from publicly available Reddit content. Although Reddit usernames were collected as part of the network construction process, these are pseudonymous identifiers used solely to construct the interaction graph and link activity records. No further private identifying information, such as email addresses or telephone numbers was collected.

In addition, user identifiers are not disclosed in the dissertation, and all users referenced in analysis tables are anonymised. Individual users are not the object of scrutiny in their own right. Rather, the analysis focuses on aggregate behavioural patterns and on the development of a methodology for identifying high-predictive users from observable interaction and forecasting behaviour. It is acknowledged that post content itself may contain personal data incidentally disclosed by users. However, content was processed solely for forecast extraction and was not stored, quoted, or attributed to individuals in the final analysis. Forecast extraction was performed using the university-provided GPT deployment, which operates under enhanced data privacy protections, mitigating the risk of third-party data retention associated with general-purpose API access. The intention of the research is therefore not to profile or expose individuals, but to investigate whether meaningful predictive signals can be inferred from public online discussion while taking reasonable steps to minimise risks to user privacy.

# 9 Conclusion

## 9.1 Summary of Findings

This project investigated whether high-predictive users in the *r/CryptoMarkets* community can be identified from their interaction patterns and behavioural attributes using graph-based machine learning. The three research questions posed in Chapter 3 are addressed in turn.

**RQ1: How effectively can language models extract and evaluate directional forecasts from informal Reddit discourse?** GPT-4o-mini achieved a macro-F1 of 0.780 on directional forecast extraction, outperforming GPT-4o (0.768) despite lower computational cost. Human audit validation supports this claim, yielding an accuracy of 0.840, macro-F1 of 0.851, and a Cohen’s kappa of 0.759, indicating substantial agreement with ground-truth annotations. Residual errors were concentrated at the forecast and non-forecast boundary rather than between bullish and bearish classes, with the model exhibiting a mild permissive bias toward extracting ambiguous content as forecasts rather than discarding it. This suggests that LLM-based extraction is reliable for well-defined directional signals but warrants stricter filtering when downstream tasks are sensitive to false-positive forecasts.

**RQ2: Do the social network interaction patterns and behavioural characteristics of a user indicate their ability to predict cryptocurrency market movements?** The analysis provides affirmative evidence. High-predictive users exhibit measurably separate behavioural profiles compared to low-predictive users in their higher comment-to-post ratios and perceived user confidence. At the network level, they display lower homophily (0.08 versus 0.93 for low-predictive users), indicating a tendency to engage across rather than within predictive groups. Community detection revealed no spatial concentration of high-predictive users within specific Louvain clusters, suggesting that predictive ability is not community-bound but is instead distributed across the network and associated with cross-group interaction patterns. GraphSAGE achieved the strongest overall performance among all the architectures, and performed better than the random baseline by 60.9% in average precision and 26.9% in AUROC, confirming that combining user activity with neighbourhood aggregation yields an advantage in identifying predictive ability.

**RQ3: Which structural and behavioural features enable graph-based models to identify users with high predictive potential, including those with limited historical activity?** Ablation analysis on GraphSAGE identified pagerank as the single most informative structural feature, with its removal reducing precision by 0.027, indicating that influence-weighted connectivity is more discriminative than raw degree. Among behavioural features, features that differentiated type of activity on the subreddit made the strongest positive contributions, reflecting the importance of en-

agement pattern over activity volume. Notably, raw degree was found to be harmful, with its removal improving precision by 0.027, consistent with the interpretation that highly active but non-predictive hub users introduce noise. GraphSAGE’s mean-aggregation over sampled neighbourhoods proved most robust under severe class imbalance and low minority-class homophily. Although the present evaluation is transductive, GraphSAGE’s inductive design offers an additional practical benefit for future deployment, where it can classify newly joined users without retraining.

## 9.2 Future Research Directions

The findings of this project suggest several concrete extensions, ranging from immediate robustness checks to broader methodological generalisations.

A natural first step is to quantify the propagation of forecast-extraction error into the final user labels, as discussed in Section 8.3.1. The most direct approach is a perturbation experiment in which a fraction of forecast labels equal to the observed GPT-4o-mini error rate (15–20%) is randomly flipped, composite scores are recomputed from the perturbed labels, and the proportion of users crossing the labelling threshold is measured. This would convert the current qualitative caveat about labelling noise into a quantitative bound on how much of the observed model performance is attributable to underlying signal rather than label artefacts.

A second direction concerns the architectural limitations exposed by the perturbation tests in Section 7.2.2, where DICE rewiring unexpectedly improved performance. This indicates that none of the evaluated architectures fully resolve the joint problem of severe class imbalance and minority-class heterophily, with class-rebalancing losses and heterophily-aware aggregation typically treated as independent design choices in the literature. Future work could explore architectures specifically designed for the joint regime, building on the components already evaluated in this work. The analysis on false negatives identified majority-class hub neighbour dilution as a systematic failure mode, motivating a modification of GraphSAGE’s neighbourhood sampling that could address this limitation by down-weighting or excluding hub neighbours whose local label distribution is dominated by the majority class. This is so that minority-class nodes are not systematically diluted during aggregation. Separately, the DICE perturbation result, in which removing intra-class edges improved test accuracy, suggests that a subset of observed edges actively obscure the minority signal. A learned edge-pruning step trained on the labelled subset could approximate this denoising automatically, rather than relying on a hand-crafted rewiring rule.

The ablation analysis in Section 7.2.1 also motivates more systematic feature selection. Although features were reduced from an initial set of 99 to 41 prior to model training, the ablation showed that removing certain features (notably raw degree) further improved precision, suggesting residual

redundancy. A formal forward or backward selection procedure, or a SHAP-based feature importance analysis run jointly across architectures, would clarify which features carry independent predictive signal and which proxy information were already encoded by graph aggregation.

Beyond the technical robustness of the present pipeline, several methodological extensions are natural. The current evaluation is restricted to a single subreddit and uses endpoint price returns. Extending the analysis across multiple cryptocurrency communities would test whether the structural and behavioural signatures of high-predictive users persist beyond *r/CryptoMarkets*, while higher-resolution price data and volatility-aware evaluation through models such as GARCH would allow forecast skill to be assessed in terms of magnitude and timing rather than direction alone. The static graph used here also limits the analysis to a snapshot of interaction patterns and a temporal graph formulation could capture how high-predictive users emerge, persist, or fade over time, enabling early identification of new informative participants as they accumulate forecast history.

More broadly, the framework itself is not specific to cryptocurrency. The combined pipeline of LLM-based extraction of evaluable claims, outcome-based user labelling, and graph-based identification of consistently informative users could be applied to any online community in which users make verifiable assertions, including technical question-and-answer platforms, prediction markets, and political discussion forums. In each of these settings, the question of who carries reliable information is increasingly difficult to answer at scale, and the methods developed here offer one principled way of doing so.

# A Appendix

## A.1 Node Feature Definitions

This appendix provides precise mathematical definitions for the 41 node features used in the graph machine learning models (Chapter 6). Let  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{W})$  denote the weighted social interaction graph,  $u \in \mathcal{V}$  a user node,  $\mathcal{N}(u)$  the set of neighbours of  $u$ ,  $w_{uv}$  the edge weight between  $u$  and  $v$ ,  $P_u$  the set of posts authored by  $u$ ,  $C_u$  the set of comments authored by  $u$ ,  $\mathcal{F}_u^P$  the set of post forecasts attributed to  $u$ , and  $\mathcal{F}_u^C$  the set of comment forecasts attributed to  $u$ . Features derived from the labelling pipeline and metadata identifiers were excluded to prevent data leakage.

### Summary Table

Table A.1 summarises all 41 features grouped by type.

Table A.1: Node features used in GNN training.

| Feature                                                                                                                                     | Description                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <i>Activity and Engagement (14)</i>                                                                                                         |                                                                            |
| num_posts, num_comments, total_activity                                                                                                     | Raw contribution volume.                                                   |
| posts_per_day, comments_per_day, activity_per_day                                                                                           | Activity rates normalised by active span.                                  |
| active_span_days                                                                                                                            | Days between first and last activity.                                      |
| comment_to_post_ratio, posts_fraction, comments_fraction                                                                                    | Balance of posting vs. commenting.                                         |
| avg_post_score, avg_comment_score, avg-upvote_ratio                                                                                         | Community reception quality.                                               |
| has_posts                                                                                                                                   | Binary indicator of posting activity.                                      |
| <i>Forecast Volume and Direction (6)</i>                                                                                                    |                                                                            |
| num_post_predictions, num_comment_predictions                                                                                               | Count of forecasts identified by GPT-4o-mini.                              |
| post_dir_frac_bullish, post_dir_frac_bearish, comment_dir_frac_bullish, comment_dir_frac_bearish                                            | Directional forecast fractions.                                            |
| <i>Forecast Confidence (4)</i>                                                                                                              |                                                                            |
| post_conf_mean, post_conf_std, comment_conf_mean, comment_conf_std                                                                          | Mean and standard deviation of GPT-4o-mini confidence scores.              |
| <i>Forecast Diversity and Horizon (10)</i>                                                                                                  |                                                                            |
| post_direction_entropy, comment_direction_entropy, post_asset_entropy, comment_asset_entropy, post_horizon_entropy, comment_horizon_entropy | Shannon entropy over forecast direction, asset, and horizon distributions. |
| post_horizon_frac_short_term, post_horizon_frac_long_term, comment_horizon_frac_short_term, comment_horizon_frac_long_term                  | Fraction of forecasts by horizon category.                                 |
| <i>Structural and Graph (7, excluded from MLP)</i>                                                                                          |                                                                            |
| degree, weighted_degree, degree centrality                                                                                                  | Local connectivity measures.                                               |
| betweenness centrality, eigenvector centrality, pagerank                                                                                    | Brokerage, recursive, and global influence.                                |
| clustering                                                                                                                                  | Neighbourhood cohesion.                                                    |

### Group 1: Activity and Engagement (Features 1–14)

1. **num\_posts**:  $|P_u|$ , total number of posts authored by  $u$ .
2. **num\_comments**:  $|C_u|$ , total number of comments authored by  $u$ .
3. **total\_activity**:  $|P_u| + |C_u|$ .
4. **comment\_to\_post\_ratio**:  $|C_u| / \max(|P_u|, 1)$ .

5. `posts_fraction`:  $|P_u| / \max(|P_u| + |C_u|, 1)$ .
6. `comments_fraction`:  $|C_u| / \max(|P_u| + |C_u|, 1)$ .
7. `avg_comment_score`:  $\frac{1}{|C_u|} \sum_{c \in C_u} s(c)$ , where  $s(c)$  is the net vote score of comment  $c$ .
8. `avg_post_score`:  $\frac{1}{|P_u|} \sum_{p \in P_u} s(p)$ , where  $s(p)$  is the net vote score of post  $p$ .
9. `avg_upvote_ratio`:  $\frac{1}{|P_u|} \sum_{p \in P_u} r(p)$ , where  $r(p) \in [0, 1]$  is the upvote ratio of post  $p$ .
10. `active_span_days`:  $T_u$ , the number of days between the timestamps of  $u$ 's first and last observed contribution.
11. `posts_per_day`:  $|P_u| / \max(T_u, 1)$ .
12. `comments_per_day`:  $|C_u| / \max(T_u, 1)$ .
13. `activity_per_day`:  $(|P_u| + |C_u|) / \max(T_u, 1)$ .
14. `has_posts`:  $\mathbf{1}[|P_u| > 0]$ , a binary indicator equal to 1 if  $u$  has authored at least one post.

#### Group 2: Forecast Volume and Direction (Features 15–20)

15. `num_post_predictions`:  $|\mathcal{F}_u^P|$
16. `num_comment_predictions`:  $|\mathcal{F}_u^C|$
17. `post_dir_frac_bullish`:  $\frac{|\{p \in \mathcal{F}_u^P : \hat{d}_p = \text{Bullish}\}|}{|\mathcal{F}_u^P|}$  where  $\hat{d}_p$  is the direction label assigned to forecast  $p$ .
18. `post_dir_frac_bearish`:  $\frac{|\{p \in \mathcal{F}_u^P : \hat{d}_p = \text{Bearish}\}|}{|\mathcal{F}_u^P|}$
19. `comment_dir_frac_bullish`:  $\frac{|\{c \in \mathcal{F}_u^C : \hat{d}_c = \text{Bullish}\}|}{|\mathcal{F}_u^C|}$
20. `comment_dir_frac_bearish`:  $\frac{|\{c \in \mathcal{F}_u^C : \hat{d}_c = \text{Bearish}\}|}{|\mathcal{F}_u^C|}$

#### Group 3: Forecast Confidence (Features 21–24)

21. `post_conf_mean`:  $\bar{c}_u^P = \frac{1}{|\mathcal{F}_u^P|} \sum_{p \in \mathcal{F}_u^P} \hat{c}_p$  where  $\hat{c}_p \in [0, 1]$  is the confidence score assigned to forecast  $p$  by GPT-4o-mini.
22. `post_conf_std`: Standard deviation of confidence scores across post forecasts,  $\sigma_u^P = \sqrt{\frac{1}{|\mathcal{F}_u^P|} \sum_{p \in \mathcal{F}_u^P} (\hat{c}_p - \bar{c}_u^P)^2}$
23. `comment_conf_mean`:  $\bar{c}_u^C = \frac{1}{|\mathcal{F}_u^C|} \sum_{c \in \mathcal{F}_u^C} \hat{c}_c$ .
24. `comment_conf_std`: Standard deviation of confidence scores across comment forecasts,  $\sigma_u^C = \sqrt{\frac{1}{|\mathcal{F}_u^C|} \sum_{c \in \mathcal{F}_u^C} (\hat{c}_c - \bar{c}_u^C)^2}$ .

#### Group 4: Forecast Diversity and Horizon (Features 25–34)

For a categorical variable  $X$  taking values in a finite set  $\mathcal{X}$ , the Shannon entropy is  $H(X) = -\sum_{x \in \mathcal{X}} \hat{p}(x) \log \hat{p}(x)$ , where  $\hat{p}(x)$  is the empirical frequency of value  $x$  across  $u$ 's forecasts, with the convention  $0 \log 0 = 0$ .

25. `post_direction_entropy`:  $H(\hat{d})$  computed over the direction labels  $\{\text{Bullish}, \text{Bearish}, \text{Non-forecast}\}$  of  $u$ 's post forecasts  $\mathcal{F}_u^P$ .
26. `comment_direction_entropy`:  $H(\hat{d})$  computed over the direction labels of  $u$ 's comment forecasts  $\mathcal{F}_u^C$ .

27. `post_asset_entropy`:  $H(\hat{a})$  computed over the cryptocurrency asset labels  $\hat{a}_p$  of  $u$ 's post forecasts.
28. `comment_asset_entropy`:  $H(\hat{a})$  computed over the asset labels of  $u$ 's comment forecasts.
29. `post_horizon_entropy`:  $H(\hat{h})$  computed over the horizon categories  $\hat{h}_p$  of  $u$ 's post forecasts.
30. `comment_horizon_entropy`:  $H(\hat{h})$  computed over the horizon categories of  $u$ 's comment forecasts.
31. `post_horizon_frac_short_term`:  $\frac{|\{p \in \mathcal{F}_u^P : \hat{h}_p = \text{short-term}\}|}{|\mathcal{F}_u^P|}$  where  $\hat{h}_p$  is the horizon label assigned to forecast  $p$ .
32. `post_horizon_frac_long_term`:  $\frac{|\{p \in \mathcal{F}_u^P : \hat{h}_p = \text{long-term}\}|}{|\mathcal{F}_u^P|}$
33. `comment_horizon_frac_short_term`:  $\frac{|\{c \in \mathcal{F}_u^C : \hat{h}_c = \text{short-term}\}|}{|\mathcal{F}_u^C|}$
34. `comment_horizon_frac_long_term`:  $\frac{|\{c \in \mathcal{F}_u^C : \hat{h}_c = \text{long-term}\}|}{|\mathcal{F}_u^C|}$

### Group 5: Structural and Graph Features (Features 35–41)

These features are excluded from the MLP baseline to enable a controlled comparison between feature-only and graph-enhanced learning. Centrality measures were introduced in Section 2.2.2.

35. `degree`:  $\deg(u) = |\mathcal{N}(u)|$ , measures the unweighted count of distinct neighbours.
36. `weighted_degree`:  $\deg_w(u) = \sum_{v \in \mathcal{N}(u)} w_{uv}$ , which is the sum of incident edge weights.
37. `clustering`:  $cc(u) = \frac{2|\{(v,w) \in \mathcal{E} : v,w \in \mathcal{N}(u)\}|}{\deg(u)(\deg(u)-1)}$  [30].
38. `degree_centrality`:  $C_D(u) = \deg(u) / (|\mathcal{V}| - 1)$ , is the size-normalised form of `degree`, dividing by  $|\mathcal{V}| - 1$  to remove sensitivity to graph size and enable comparison across networks.
39. `betweenness_centrality`:  $B(u) = \sum_{s \neq u \neq t} \frac{\sigma(s,t|u)}{\sigma(s,t)}$  where  $\sigma(s,t)$  is the total number of shortest paths from  $s$  to  $t$ , and  $\sigma(s,t|u)$  is the number of those paths that pass through user  $u$  [49].
40. `eigenvector_centrality`: Principal eigenvector component  $x_u$  satisfying  $\mathbf{Ax} = \lambda_{\max} \mathbf{x}$  [20].
41. `pagerank`: Stationary distribution score of a random walk who, at each step, either follows an edge with probability  $d$  or jumps to a random node with probability  $1 - d$ . The score is defined as:  $\text{PR}(v) = \frac{1-d}{n} + d \sum_{u \in \mathcal{N}(v)} \frac{\text{PR}(u)}{\deg(u)}$ , where  $d \in [0, 1]$  is the damping factor (typically set to 0.85),  $n$  is the total number of nodes, and  $\deg(u)$  is the out-degree of user  $u$ . [50]. Stronger edge weights contribute more to a node's score.

## A.2 Heuristic Labelling Specification

This appendix details the rule-based baseline referenced in Section 4.3.3. The heuristic was implemented prior to the LLM pipeline as a scalable alternative to manual annotation, but was found to be unreliable on informal Reddit discourse and was therefore replaced.

**Direction** was determined by token frequency scoring over three fixed lexicons applied to lowercased post and comment text:

- **Bullish terms:** buy, bullish, long, accumulate, moon
- **Bearish terms:** sell, bearish, short, dump, crash
- **Scam exclusion filter:** guaranteed, 100x, telegram, dm me

A post was labelled *Bullish* or *Bearish* according to which lexicon yielded the higher token count, and excluded entirely from the forecast set if any scam-filter term was present. Ties and zero-match cases were classified as *Neutral* and hence *Non-forecast*.

**Time Horizon** were assigned via sequential keyword matching across short-, mid-, and long-term categories. Posts containing short-term markers (e.g. *today*, *tomorrow*, *this week*) were assigned the short-term category and remaining posts were checked against long-term markers (e.g. *next year*, *by 2030*); all others defaulted to mid-term.

**Assets** were extracted via case-insensitive substring matching against a fixed symbol map covering canonical tickers (BTC, ETH, SOL, XRP, etc.) and their common name variants (*Bitcoin*, *Ethereum*, etc.).

### Failure Modes

The heuristic produced unreliable labels for three principal reasons. The use of hedging through phrases such as *might moon* or *could dump* satisfied lexicon matches but actual forecasts do not have genuine directional conviction. Additionally, sarcasm was common, but it inverted directional intent (*yeah right*, *BTC will moon*) which was not detected by token-level matching. Temporal contexts are also often implicit, and the absence of explicit horizon markers caused the mid-term default to be applied indiscriminately, regardless of whether the original post implied a specific timeframe. These limitations motivated the adoption of LLM-based extraction (Section 4.4), which handles pragmatic context, hedging, and implicit temporal cues that token-level rules cannot capture.

## A.3 Representative Confidence Score Examples

Table A.2 presents paraphrased examples drawn from the canonical year-long dataset of 125,170 posts and comments, illustrating the qualitative differences in perceived confidence scores in Section 4.4.3.

Table A.2: Representative and paraphrased examples of high and low perceived confidence forecasts extracted by GPT-4o-mini. Confidence scores are on a [0,1] scale.

| Conf. | Paraphrased post excerpt                                                                                                              | Direction | Asset    |
|-------|---------------------------------------------------------------------------------------------------------------------------------------|-----------|----------|
| 0.9   | User expresses strong conviction that a specific altcoin will lose all value and advises others to avoid it entirely.                 | Bearish   | Alt-coin |
| 0.8   | User predicts a notable price increase for XRP and notes that ADA is gaining renewed media attention and popularity.                  | Bullish   | XRP, ADA |
| 0.6   | User suggests that if historical market cycles repeat, Bitcoin could reach a peak price around \$180k by late 2025.                   | Bullish   | BTC      |
| 0.5   | User acknowledges that monthly returns are often negative but argues that sustained accumulation will eventually yield profit.        | Bullish   | BTC      |
| 0.5   | User solicits community opinions on their portfolio allocation strategy ahead of the new year, without expressing a directional view. | Neutral   | —        |

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